CONCEPT FOR THE IDEAL RETRO COMPUTER · FINAL OVERVIEW

Neo 500, 2000 and 4000

Three cases, one motherboard in full configuration, the back panel and Neo OS at work. All drawings are original concepts: a fast ARM as the sole master of the system, a strong FPGA fabric for retro cores and, in the larger models, the Satellite x86 coprocessor.



Dijana, the Neo OS mascot

Architecture Models Board Ports Neo OS Apps in the box Personality Retro and Future Shock Beauty Sound Even easier For people Files Workshop Runs everything From the Unix world Themes and voice Undo and transfer Best of the past Principles ROM and cartridge Perception Compression Processor and threads Upgrades Memory Camera Energy Applications Games Arcade Together Co-op Peripherals People's computer Neo 100 and Revive Knowledge Smart solutions Standard Hardware standard Prices Comparison

ARCHITECTURE

How Neo is put together

One master of the system, dedicated chips around it and one shared memory. Nothing works as a separate machine fighting for control.

Processor

ARM leads, the FPGA changes shape

MAIN CPU

A fast ARM runs Neo OS and translates x86 programs (FEX/Box64 under Bridge).

FPGA + CGRA

Chameleon loads retro cores (Z80, 6502, 68030) and accelerators.

SATELLITE X86

A separate x86 coprocessor for Windows, DOS and drivers, like the Bridgeboard card of old.

Chipset

Modern custom chips

GRAPHICS

VSA-6000 (3D chiplets), Ray (ray tracing and blitter), Panorama (up to six displays, layers, AV1 encoder).

OTHERS

Newton (physics), Orpheus (sound), Copernicus (Copper 2), Neuron (AI), Courier (I/O), Chronos (state snapshot).

The future

One Neo package

CHIPLETS

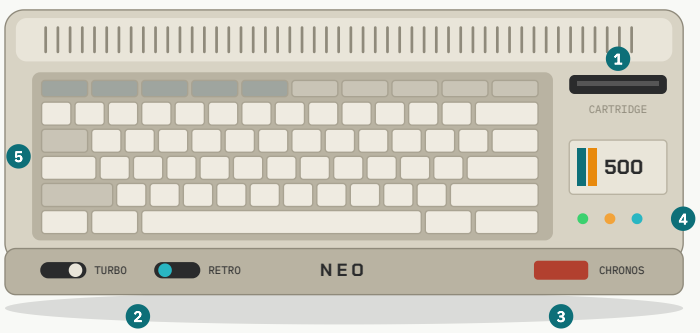
ARM, FPGA, VSA-6000 and an optional x86 in one package, connected over UCIe, with HBM memory.

MEMORY

Coherent (CXL), so all parts see the same data without copying.

MODELS

The family of cases



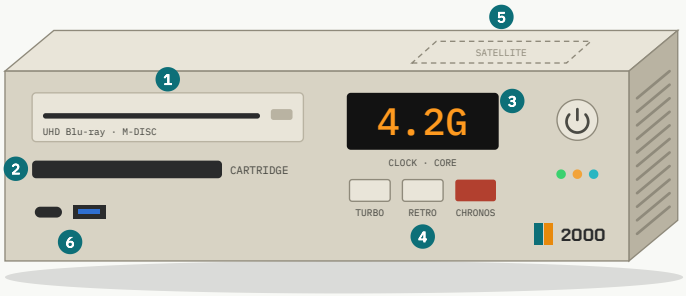
ENTRY MODEL

Neo 500

A computer in a keyboard, without the x86 part. A fast ARM runs Neo OS and translates x86 programs, and the FPGA loads one retro core at a time through Chameleon.

1. Cartridge slot for games, cores and ROM packs
2. Turbo and Retro switches: full clock or the exact clock of the old system
3. Chronos button: snapshot of the whole machine state and return
4. LEDs: power, disk, active FPGA core
5. Mechanical keyboard with a separate function row

Back: 2× DE-9, VGA, HDMI, 2× USB-C, USB-A. Underneath, a cover for M.2 and memory expansion.

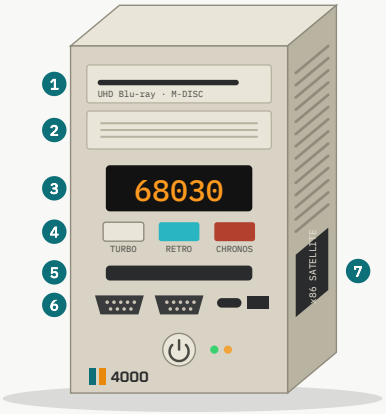


MID-RANGE MODEL

Neo 2000

A desktop case that the monitor sits on. Satellite x86 is an optional module inserted from the top, so the same computer can be bought without it and upgraded later.

1. Bay for a UHD Blu-ray or M-DISC drive (option)
2. Cartridge slot
3. LED display: clock speed in Turbo mode, core name in Retro mode
4. Turbo, Retro and Chronos buttons
5. Cover for the Satellite x86 module (COM type)
6. Front USB-C and USB-A



TOP MODEL

Neo 4000

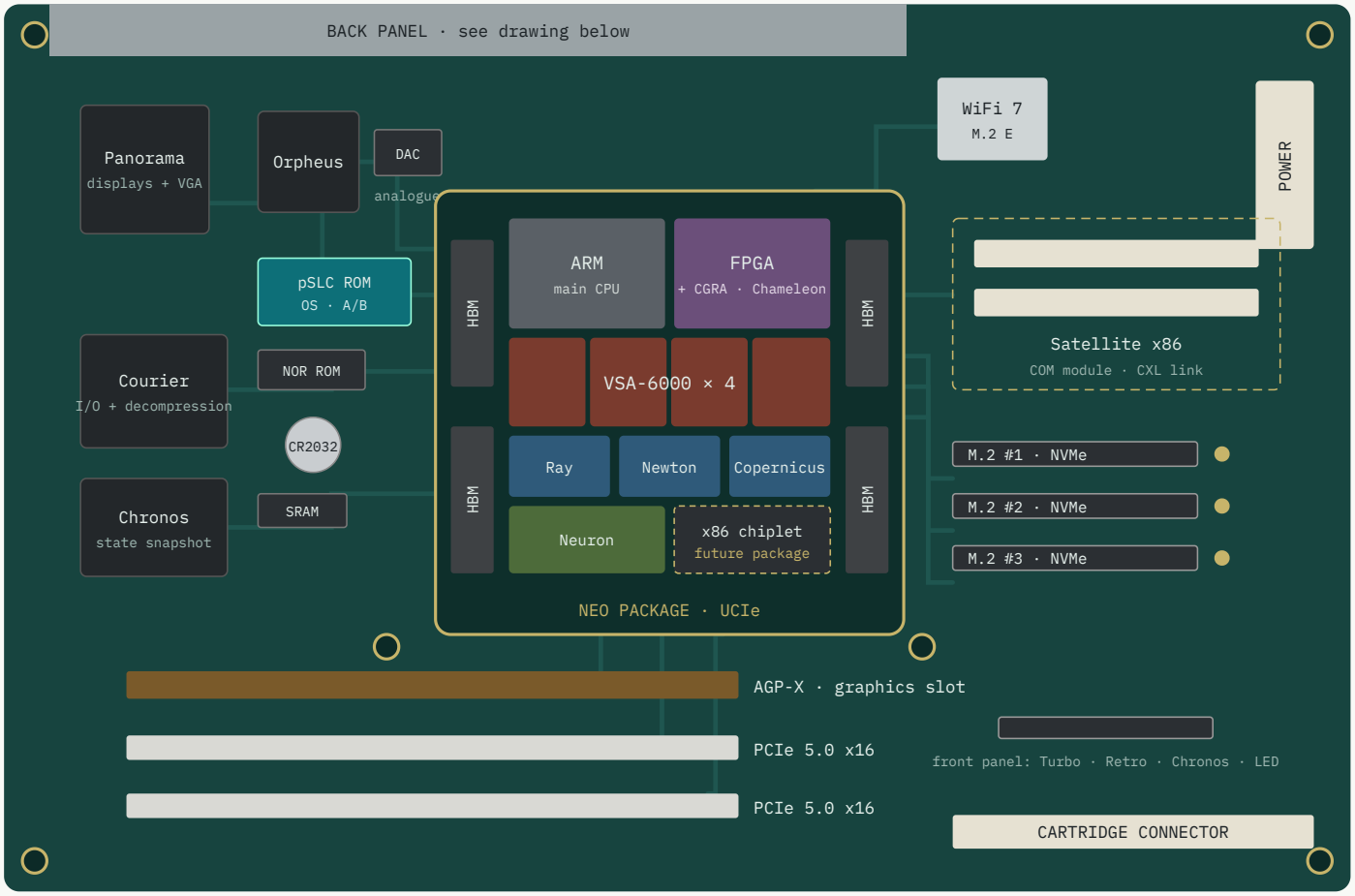
A tower with a built-in Satellite x86 coprocessor, the full VSA-6000 configuration and all the slots: AGP-X, two PCIe 5 and three M.2. In Retro mode the display shows the name of the loaded core.

1. UHD Blu-ray / M-DISC bay
2. Free 5.25" bay
3. LED display (here: loaded core 68030)
4. Turbo, Retro and Chronos buttons
5. Cartridge slot
6. Two front DE-9 joystick ports, USB-C, USB-A
7. Side plate: Satellite x86 built in

INSIDE

Neo 4000 motherboard

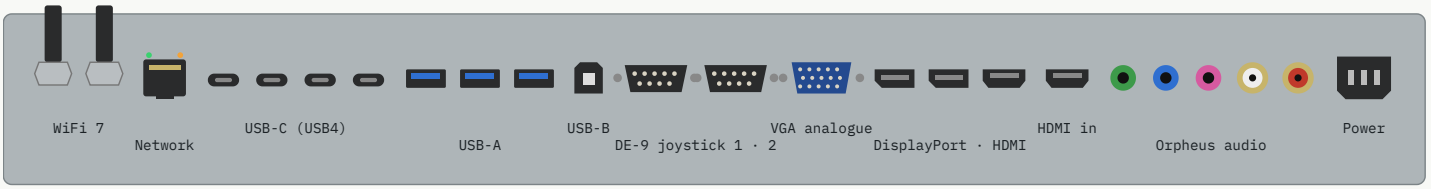
Full configuration. In the middle is the Neo package: chiplets connected over UCIe, with HBM memory in the same chip package. Neo 500 and 2000 use the same board with fewer slots populated.



PORTS

Back panel

The same layout on Neo 2000 and 4000. Analogue VGA goes through a quality RAMDAC output from Panorama, in the spirit of the best graphics cards of the late nineties.

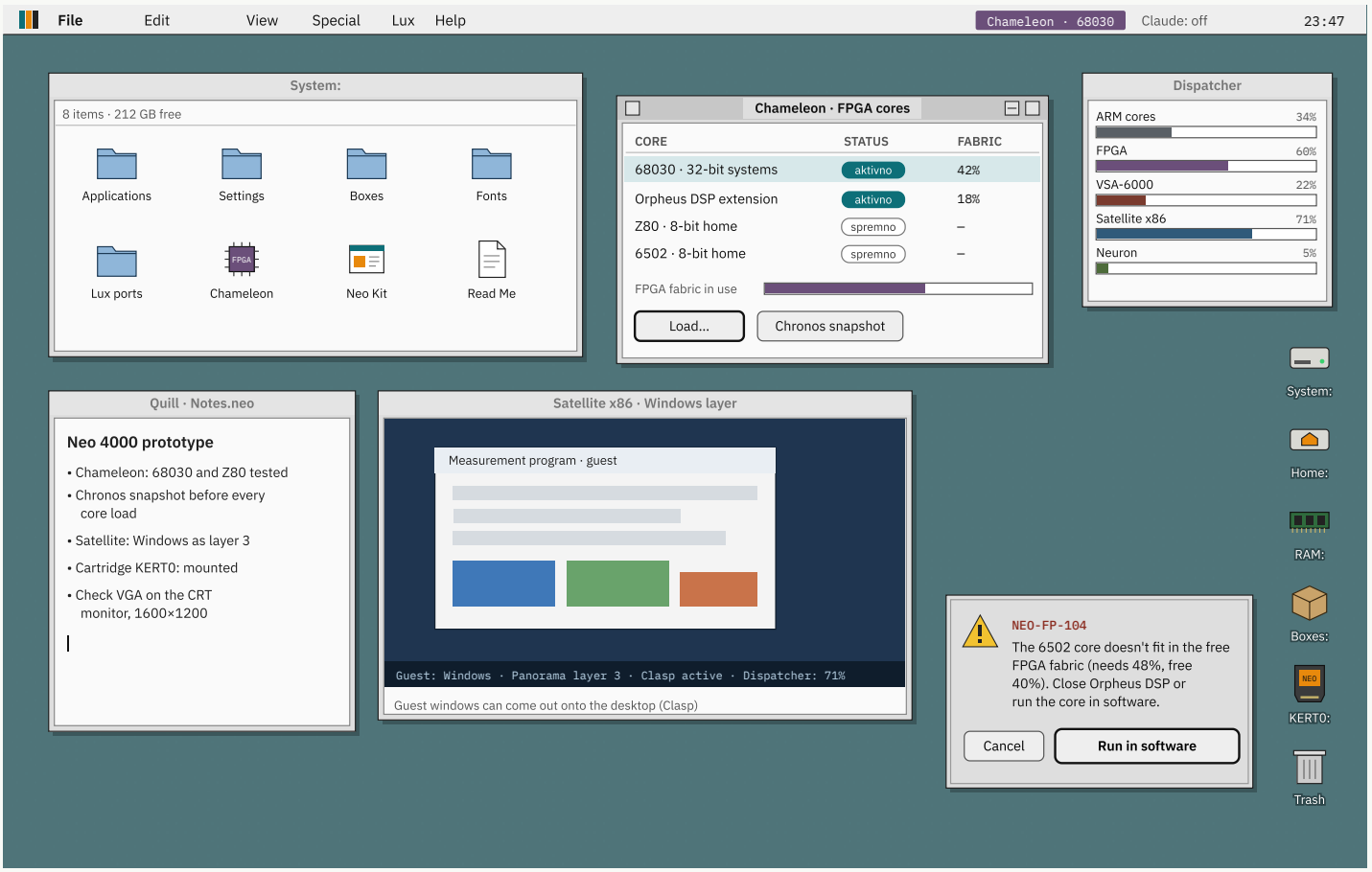


Neo 2000 / 4000 · rear view

SOFTWARE

Neo OS at work

The Neo Platinum look. On the desktop are the logical volumes (System:, Home:, RAM:, Boxes:) and an inserted cartridge. Chameleon holds the 68030 core in the FPGA, Satellite shows Windows as a Panorama layer, and the error message carries a code and a suggested fix.



EVERYTHING IN THE BOX

What Neo OS has from the first power-on

One application for every everyday need, no more and no less. They all have an Easy / Pro switch in the same place: Easy shows a few essential options, and Pro opens the full palette. The rest (3D, CAD, professional DAW) is in the Treasury, one click away.

Need	Application	What it does	Engine
Reading	**Leaf**	PDF, EPUB, FB2, comics (CBZ, CBR), DjVu. Public libraries via OPDS, night mode, notes in the margins, reading aloud via Neuron.	Poppler, MuPDF, Foliate
Files	**Cabinet**	Two panels, buttons the user makes, batch renaming, preview without opening, archives, FTP and SFTP as ordinary folders.	libarchive, GVfs

Need	Application	What it does	Engine
Pictures	**Album**	Fast browsing of thousands of pictures, faces and places recognized locally, easy touch-ups, comparing two pictures.	libvips, darktable
Sound	**Wave**	Waveform and multitrack mode, noise removal, mastering, recording from the Orpheus input.	Audacity engine, LV2
Music	**Turntable**	Library, radio, podcasts, Cyrillic shown correctly in titles.	GStreamer
Video	**Cinema**	Plays every format, subtitles are found and translated locally.	mpv
Planner	**Planner**	Notes, calendar, contacts and reminders in one place.	CalDAV, CardDAV
Maps	**Atlas**	Maps that work without the internet, routes on foot, by bike and by car.	OpenStreetMap
Chat	**Chat**	Encrypted messages and video calls, without a phone number.	Matrix, Jitsi
Passwords	**Keykeeper**	Passwords and keys are kept in the Safe chip, never in the cloud without your consent.	KeePassXC format
Scanner	**Scanner**	Scanning and text recognition, Serbian Latin and Cyrillic.	Tesseract
Translation	**Interpreter**	Translating text and pages without sending data to the internet.	Bergamot, Neuron
Phone	**Link**	Files, messages and notifications between phone and computer.	KDE Connect
Retro	**Arcade**	Library of games and cores, cartridges and floppies in one place.	Chameleon, RetroArch cores

Enables

Teaches, doesn't dumb down

WHY?

Every setting has a button that explains in plain language what it does and what will change.

RECORD STEPS

Any mouse action can be recorded as a Lux script and replayed, so you learn automation along the way.

Doesn't spy

No account, no telemetry

ACCOUNT

The computer works without any account. An account is needed only to buy in the Treasury.

DATA

No ads, no telemetry. An error report is sent only when you send it, and you see what it contains.

Stays out of the way

No fear of mistakes

UNDO

Undo in every application, and Chronos returns the whole system to yesterday's state.

PEACE

No pop-up offers and no updates in the middle of work. Renewal asks once and waits.

PERSONALITY AND SIMPLICITY

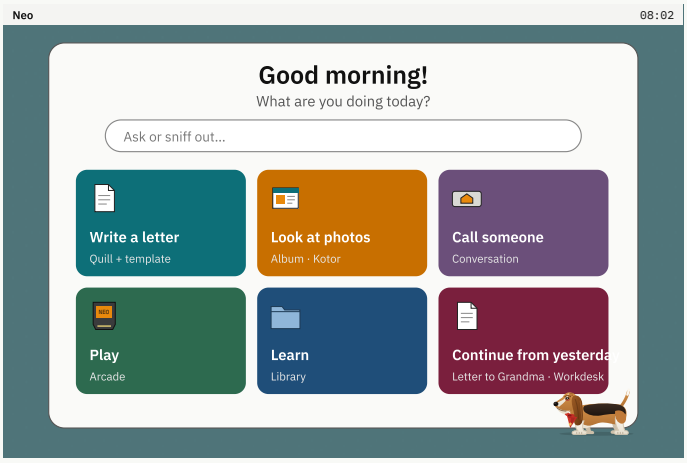
A computer with a soul that stays out of your way

Neo should be liked at first sight, learned in an afternoon and give results right away. That is why it speaks everyday language, has a resident named Dijana and takes what is simplest in mobile apps. A lesson from the past: Microsoft Bob turned the whole system into a cartoon house, and Clippy kept interrupting. Neo takes only the everyday words and a personality that doesn't impose itself.

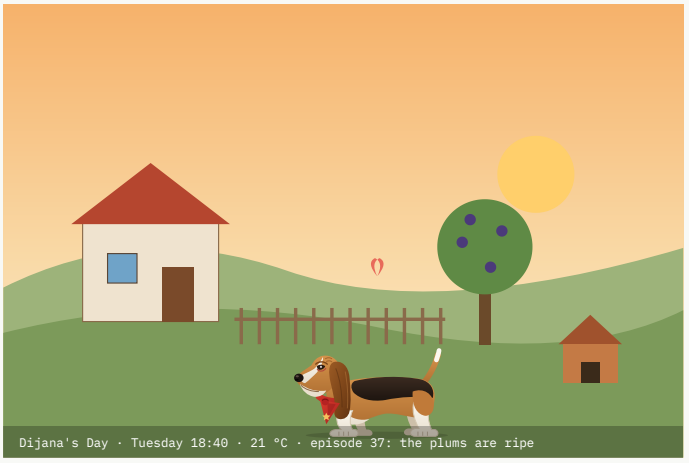
Neo desk: shelves, a search that sniffs and Dijana by the Bin



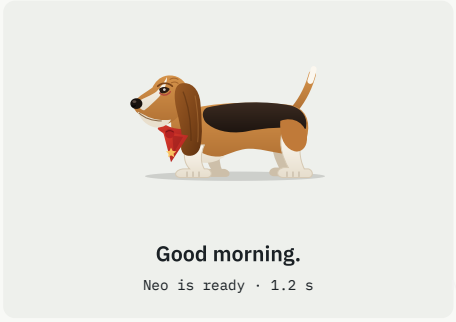
Start screen: "What are you doing today?"

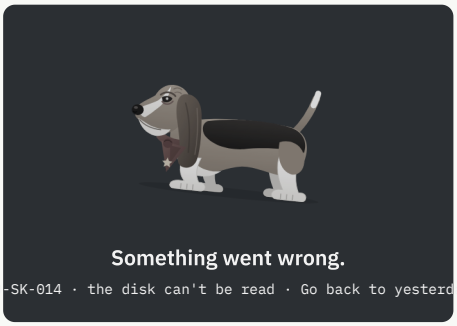


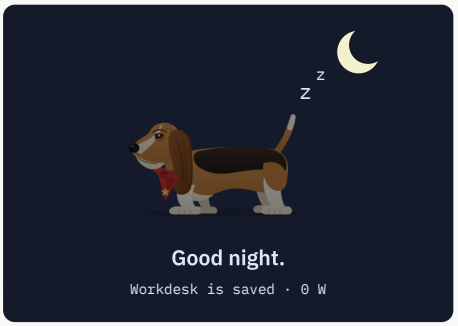
Screen saver: Dijana's Day



System personality: power-on, error and shutdown







Easy (everyday)	Pro (technical)	Why
Desk	Desktop	On the desk is what you are working on right now.
Folder	Folder, directory	Just like a paper folder.
Mailbox	Downloads	Everything that comes from outside arrives here.
Bin	Trash	Can be brought back until it is emptied.
Foundation	System:	Everything stands on it and nobody touches it.

Easy (everyday)	Pro (technical)	Why
Toolshed	Apps:	All programs in one place.
Guest rooms	Boxes:	Windows and other guests live separately.
Notepad	RAM:	Quick notes that get erased.
Templates	Blocks:	Ready-made beginnings for a letter, an invoice or a project.
Workdesk	RAD:	Stays the way you left it, even after a restart.
Caretaker	Dispatcher	Knows who does what in the house and what uses power.
Keys	Permissions	Who may enter which room.
Neighbourhood / World	Local network / internet	Nearby computers and everything else.
Household / Kids' room	Profiles / child profile	Everyone has their own place.
Back to yesterday	Chronos rollback	Says exactly what it does.
Nap / Good night	Sleep / shutdown	Just like people.
Shelves	Dock, launcher	You put on the shelf what you need at hand.

Dijana

A living resident, not a pushy assistant

WHERE SHE LIVES

She sleeps in the corner of the menu bar. When she has a useful tip, she just raises her ears. Click if you want, otherwise she keeps sleeping.

WHEN SHE SPEAKS UP

At first power-on, after an error (to explain what happened) and when you call her by voice or with a key. Never in the middle of typing.

LEARNS

Every tip only once. “I don't need this” turns off that whole kind of tip, and after being ignored three times she goes quiet on her own. She never sells anything.

Sniffing

“Dijana, sniff out…”

SEARCH

A basset is a scent hound, so search sniffs: Dijana lowers her nose and follows the trail, and the results come up as you type.

LOST

For a lost file she says where and when it was last seen: “It was in the Mailbox yesterday at 18:20, then you moved it to Work.”

Humour

Trying to eat a document

WHEN

Rarely: when you drag a document to the Bin, she sometimes catches it in her teeth and asks “May I?”, then drops it and wags her tail.

LIMITS

She never really touches the document, never in the middle of work and never more than once a day. Jokes are turned off with one switch.

On the desk

A small presence that doesn't get in the way

BEHAVIOUR

Sometimes she walks along the shelf, lies down next to a program that runs for a long time and wags her tail when Guardian has finished a backup. When the Sun battery runs low, she lies down in the shade.

PEACE

While you type or watch a film, she doesn't move. On battery she isn't animated, so as not to use power.

Screen saver

Dijana's Day

WHAT IT IS

A living story in episodes: Dijana in the backyard, at the market and by the river. Time of day and weather follow the real weather outside, and holidays and your calendar change the scene.

EPISODES

The story continues where it stopped. It is assembled from pieces on Neuron, so it rarely repeats. Our own original story and drawing.

System personality

Like the smiling Mac, only ours

STARTING UP

Dijana wakes up, stretches and raises her ears: everything is fine.

ERROR

Sad Dijana with lowered ears, an error code and the solution in plain language.

SHUTDOWN

“Good night”, and Dijana curls up. Along with soft sounds from Neo sound and animations that can be turned off.

Residents

Mascots of your choice, the only shop in the system

WHAT THEY ARE

A resident replaces or complements Dijana: a panda, bear, snake, cat, hedgehog, stork or a cartoon character. Each has the same behaviour (sleeps, sniffs during search, jokes, rejoices and mourns), its own screen saver story and its own sounds.

BUYING

In the Treasury, once and forever, without a subscription. The author gets most of the revenue, so local illustrators and animators get a market. Free community residents come under a free licence.

LICENCES

Well-known characters only with the official licence of the rights holder (cartoons, comics, games). First candidate for a deal: the snake from the song “Ide zmija” (“Here Comes the Snake”) by DJ Nick DeeTronic, as a dancing resident that dances to the Turntable. No unofficial copies and no real people. A country can add a national character, for example an animal or a fairy-tale character.

SAFETY

A resident only receives events (“search started”, “backup done”). It sees neither files nor the network, sells nothing and shows no ads.

Personal

A computer that is only yours

NAME AND GREETING

The computer gets a name, and the resident greets you by yours.

LOOK

One accent colour throughout the system, and the background. Neo checks the contrast, so everything stays readable.

SOUND AND VOICE

You choose the power-on sound or compose your own in the Orpheus synthesizer. You choose the voice and how much Neo talks.

NOOK

A small place on the desk for a picture, a quote or the Flowerpot, a plant that grows as you use the computer and never wilts.

ANNIVERSARY

“Today it's been a year since we've been together”, and a short look back at what you made. Everything stays on the computer.

Shelves

Instead of a Start menu

HOW

As many shelves as you want, along any edge. Anything goes on them: programs, documents, folders, Lux scripts and live tiles (clock, mixer, Workdesk, Sun).

SEPARATE

Shelves are for launching, and what is running is in the Caretaker in the menu bar. Each task has its own shelves. For everything else there is the command palette.

Helpers

Instead of Commodities

WHAT THEY ARE

Small background tools: hotkeys, bring window to front on click, focus follows mouse, abbreviation expansion, automatic actions.

SAFE

One panel turns them on and off. A helper that sees the keyboard or mouse must request it through Keys.

Like a phone

As simple as a good mobile app

SCREEN

One main action per screen, a big button in the same place. Search is always at the top, and Easy/Pro hides the rest.

NO FEAR

Instead of “Are you sure?” the action is carried out, and “Undo” waits at the bottom. There is no Save button, everything saves itself.

EMPTY

An empty screen isn't empty: it says what is done here and offers a “Start from…” template.

Task-based

“What are you doing today?”

GETTING STARTED

The first screen offers big tiles: Write a letter, Look at pictures, Call someone, Play, Learn. Each one opens a ready-made task with the right programs and shelf.

LEARNING

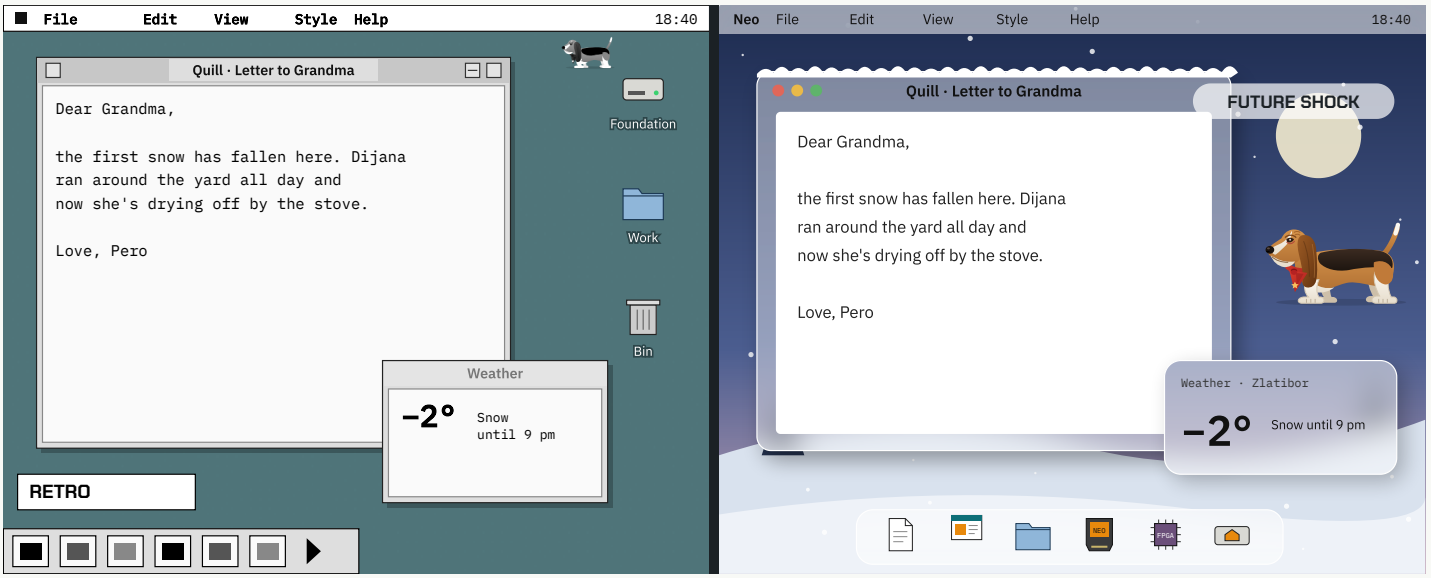
First steps through a game in 15 minutes, and every program shows three things the first time, no more.

TWO FACES OF THE SYSTEM

Retro and Future Shock: the same app, two worlds (and a third in preparation)

Neo has two global styles. Retro is a clean look in the spirit of the classic Mac systems of the nineties: grey silver, striped title bars, crisp letters and not a single effect. Future Shock is everything from the Beauty section without the burden: glass, tiles, physics, 3D and the seasons. Every app runs in both without a single extra line of code, and switching takes one click.

The same Quill app, the same document: Retro on the left, Future Shock on the right



Part of the system	Retro	Future Shock
Window	Grey frame, striped title bar, square buttons, hard shadow	Frosted glass, rounded corners, soft shadow following the time of day
Menu	White bar with black letters, drop-down menus without transparency	Glass menu with the beauty level and a clock
Shelves	Control strip along the edge of the screen, icons without motion	Glass shelf, icons bounce, snow settles
Window overview	List of windows in the application menu	Flip Through in a 3D arc
Tiles	Small add-ons in their own little window, like desk accessories of old	Live glass tiles
Bin	The Bin bulges when it is full	The document crumples up and flies off in an arc
Type	Cube, our crisp screen typeface, at size 12	IBM Plex Sans with smooth edges
Icons	32 × 32, limited palette, black-and-white if you like	Full colour with depth and shadow
Sound	Basic or Chip theme	Crystal or Wood theme, with spatial sound
Dijana	A small black-and-white pixel Dijana in the corner of the menu	Full Dijana who walks around, shakes off snow and chases paper
Motion	Just an outline that expands when a window opens	Physics, bounce and motion blur

One codebase

The app says what, the style says how

NEO KIT

The app describes a button, list, field and toolbar by meaning, not by appearance. Panorama draws the style, so the same app gets both faces without changes.

ITS OWN DRAWING

Apps that draw themselves (games, Sculptor, Arcade) keep their own look, and the frame around them follows the style.

GUESTS

Windows, Linux and Android programs get a frame and menu in the chosen style, and stay as they are inside.

Switching

One click, no restart

INSTANTLY

The style changes with a shortcut or from the menu, instantly in all open windows, without losing any work.

PER APP

Each app can have an exception: for example Cabinet in the Retro style and the rest in Future Shock.

POWER

On battery Neo suggests Retro with a single message, because Retro uses almost nothing for graphics.

Combinations

Style, colours and level are separate

COLOURS

Silver, Slate, Contrast and Phosphor work in both styles. Retro with Phosphor looks like an old green monitor, and Future Shock with Slate like a city at night.

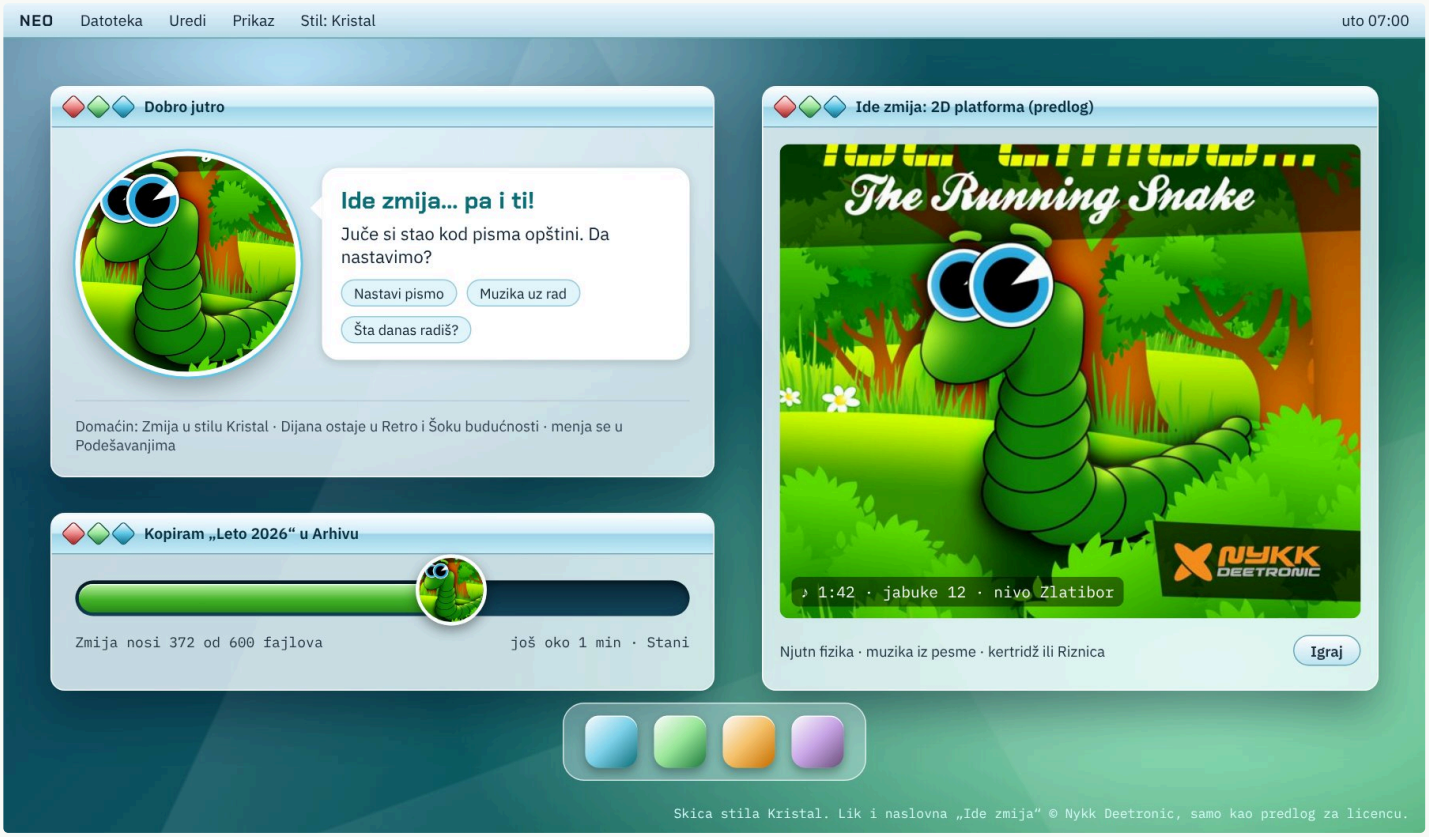
LEVELS

Calm, Lively and Lavish apply within Future Shock. Retro is always calm.

ACCESSIBILITY

Contrast, magnification and “Less motion” work the same in both styles.

Sketch of the Crystal style: the Snake welcomes you, carries files and has its own game



Character and title “Ide zmija” © Nykk Deetronic, shown only as a licensing proposal.
The crystal frames are our sketch until the author's theme arrives.

Crystal · proposal

A third style with the Snake as host, under the author's licence

WHAT

A third global style in the spirit of the Crystal theme that DJ Nick (Nikola Tomić, Deetronic Design Studio) made for AmigaOS 4: crystal window frames, its own pointer, icons and backgrounds. As with the other two styles, every app works in Crystal too without code changes.

HOST

In Crystal the host is not Dijana but the Snake from the song “Ide zmija”: it welcomes you at power-on, dances to the Turntable and carries files along the progress bar. Dijana remains the host in Retro and Future Shock.

SOUND

The author's power-on and error sounds, together with our Crystal sound theme.

GAME

“Ide zmija”, his running game, is ported to Neo as a 2D platformer with Newton physics and the song's music, in Neo games and on cartridge.

LICENCE

One contract with the Deetronic studio for the style, sound, host, resident and game, with author credit and a share of Treasury sales. Until permission is signed this is only a proposal.

For developers

Checking in both styles is mandatory

TOOL

Neo Studio shows the app window in both styles side by side while you build it, and warns when something doesn't fit or loses contrast.

TREASURY

An app passes an automatic check in both styles and both colour schemes before publication. Whoever draws their own controls must provide both versions.

NAME

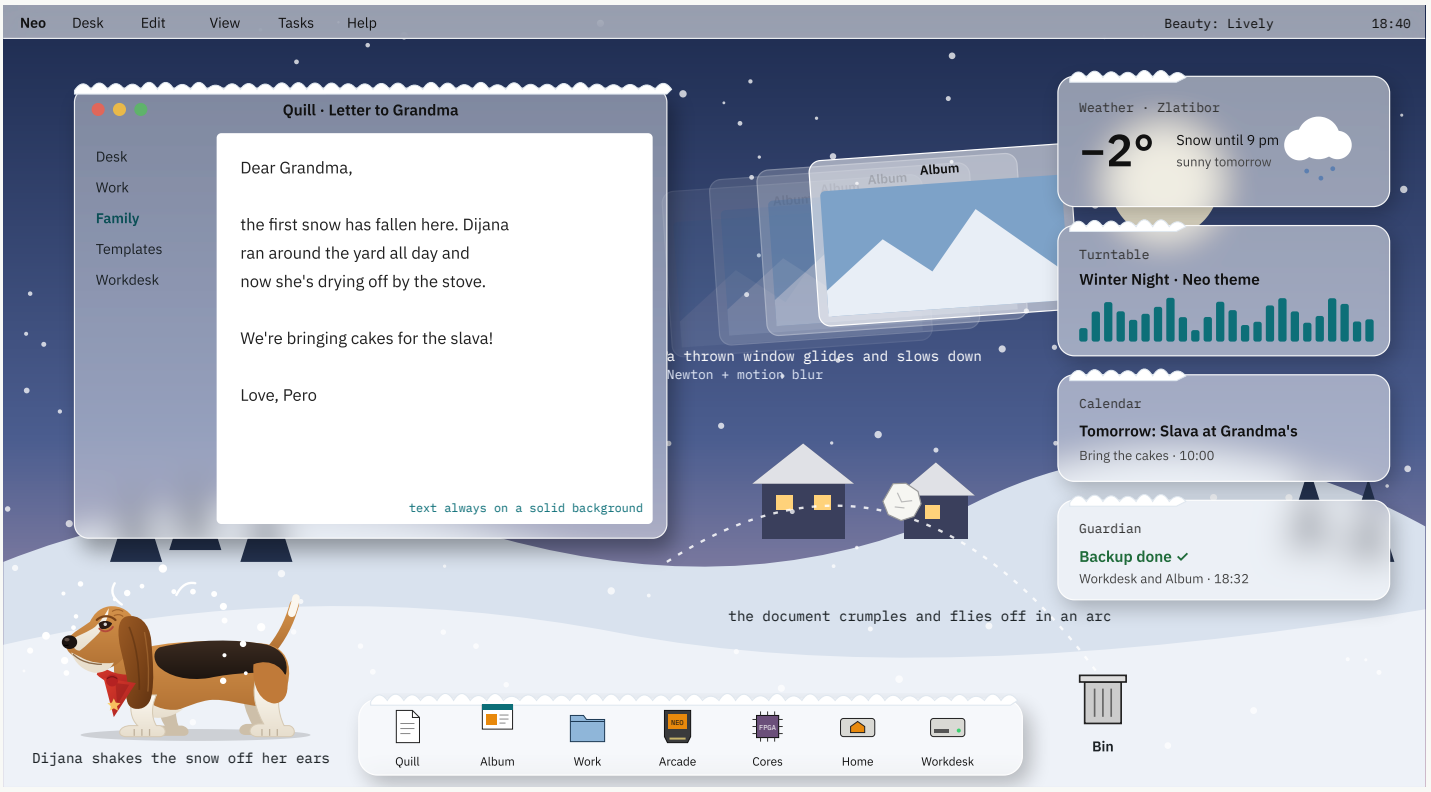
“Future Shock” is a nod to the famous book about how people experience rapid change. That is exactly why Retro exists too: whoever doesn't want change doesn't have to put up with it.

BEAUTY WITHOUT THE BURDEN

Glass, tiles and physics, while the processor rests

The best from the era when computers loved to show off: tiles with live data, transparent glass, a 3D window overview and physics like in games. The difference is that on Neo all of this is done by dedicated chips (Ray for 2D and glass, Newton for physics, VSA-6000 for 3D, Panorama for screen layers), so the main processor and the battery barely feel it. Nothing slows down work, and everything turns off with one switch.

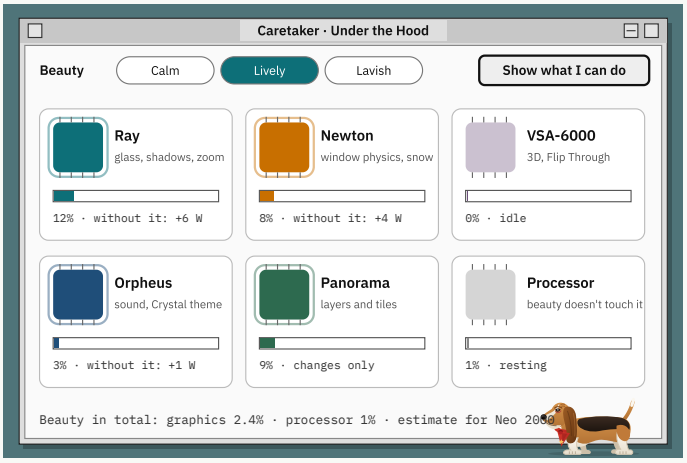
A winter evening on the Neo desk: glass, tiles, snow on the edges and Dijana shaking herself off



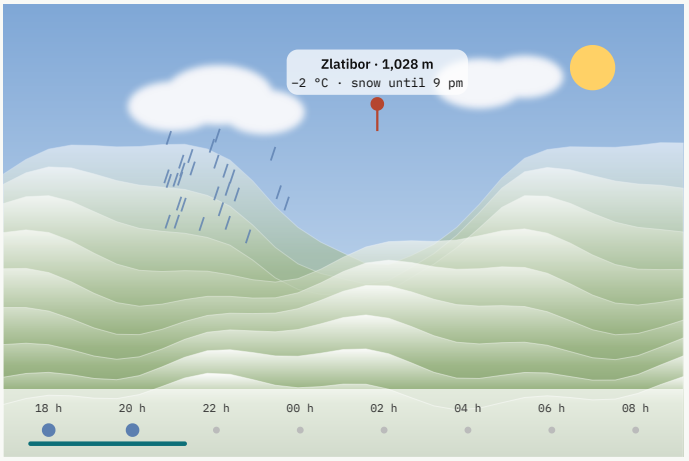
Flip Through: all windows in 3D, in one move



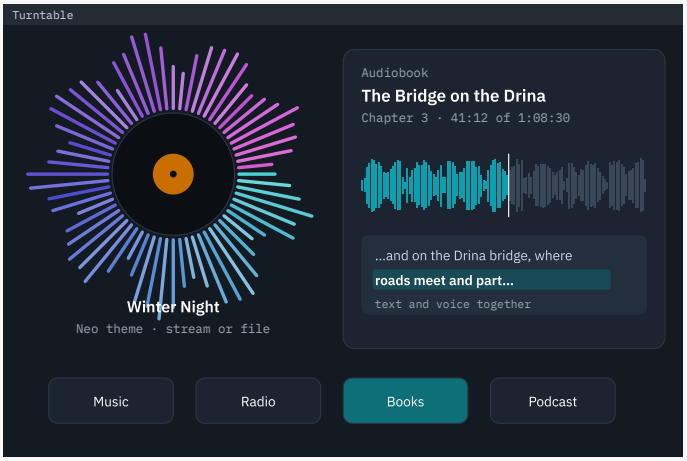
Under the Hood: which chip is working now, while the processor rests



The forecast as a live scene over Atlas terrain



Turntable: music, streams and audiobooks you can see



Tiles

Live data on the desk, the shelf or in the Nook

WHAT THEY ARE

Small live cards: weather, calendar, Turntable, exchange rate, Sun station, Guardian backups, a note, a clock for another city. The idea of the Vista-era sidebar, but without the slowness.

HOW THEY WORK

Each tile is a separate layer in Panorama and refreshes only when its data changes, at most once a second. When you can't see it, it does nothing.

SAFETY

A tile gets only the data it displays, through Keys. It cannot read files or send data without permission. They are made in Neo Kit in an afternoon.

Glass

Transparent, but always readable

WHERE

The menu bar, window titles, sidebars, Shelves and tiles are made of frosted glass through which the background shows. Text and documents are always on a solid, opaque backing.

CONTRAST

Ray measures the background under the glass and increases the blur or darkening by itself, so text never drops below the required contrast. With the Contrast theme the glass becomes solid.

LIGHT

Window shadows follow the time of day: short and cool in the morning, long and warm in the evening. The reflection on the glass shifts slightly when you move a window.

Newton

Physics that helps, not just entertains

WINDOWS

A thrown window glides and slows down, then bounces off the edge of the screen or moves to another monitor. Windows spread apart slightly so they don't overlap when there are many.

BIN

A document crumples up and flies in an arc into the Bin. The Bin fills visibly, and emptying is a short shake.

SHELVES

Icons on the shelves move aside when you drag something, and a program that is starting bounces gently. A folder dragged over another folder springs open.

SEASONS

In winter snow falls and settles on window edges and the shelf, in autumn leaves, in summer plums from Dijana's backyard. Move a window and the snow falls off. Only at the Lavish level, and never over a document.

VSA-6000

3D capabilities in everyday work

PRELISTAJ

In one move all windows fit into a 3D arc, with depth of field: the active one is sharp, the others slightly blurred.

ACCUMULATION EFFECTS

The spirit of the old T-buffer: motion blur on a thrown window, soft shadows, depth of field and perfectly smooth letter edges, all in one pass on the GPU.

ZOOM

Maps, pictures and PDFs zoom smoothly and without blur, because they are drawn from vectors and tiles at full resolution. The magnifier for the visually impaired works the same way.

RETRO

Glide mode and the Voodoo core in Arcade for old 3D games, with a sharper image than back then.

Every day

Where 3D helps an ordinary user

FORECAST

Weather as a live scene: clouds, rain and sun over the Atlas terrain of your area, with a clock across the whole day. The same scene can also be the desk background.

SOUND

Turntable shows music and streams as a gentle visualization, and an audiobook as a voice waveform with chapters. A calm story, a calm picture.

PICTURES AND VIDEO

Album as a light table with pictures that stack and turn. Video calls with a blurred background and light correction on Neuron.

READING

Leaf turns a page like paper, and a comic leads you panel by panel.

Revived

Apps that others threw out but people loved

GAMES

Cosmos Pinball with real ball physics, Grandmaster 3D with glass pieces, Solitaire and Minesweeper with cards that cascade at the end of the game.

TOOLS

Simple film editing from Album (title, music, transitions) in five minutes, a live background (Dijana's Day on the desk) and tiles.

RULE

They are all our original apps with our own names. We take the idea, not the code or the look.

Residents

Physics and residents together

DIJANA

She shakes snow off her ears, runs after a crumpled paper and brings it back, and lies down on the Sun station tile when it's warm.

OTHERS

The panda catches falling leaves, and the snake dances to the rhythm of the music from Turntable, because Orpheus detects the beat.

LIMIT

A resident gets only physics and beat events. It never covers a document and doesn't move while you type.

Flip Through

A 3D switcher that is also practical

LESSON

The old 3D switcher with windows stacked like cards looked great, but the windows at the back were hidden and had no names. Neo takes only what helps.

LAYOUT

Windows fit into an arc, all visible and named, grouped by Tasks. They stay live: video plays, a copy progresses.

FAST

Typing filters (“let” leaves Letter), the chosen window flies exactly into its place so the eye doesn't lose track, and it all takes under 150 ms. It works across all monitors, and the ordinary window list stays for those who like it.

DEPTH ON A FLAT SCREEN

Good 3D on a 2D surface: slight background parallax when the mouse moves, soft shadows that show what is on top and layered glass. Depth of field only in Flip Through.

Show what I can do

Hardware power anyone can understand

UNDER THE HOOD

A window in the Caretaker with a picture of the chips that light up while they work: Ray draws glass, Newton computes snow, Orpheus plays, VSA-6000 drives 3D. The main processor shows that it is resting, and next to each chip it says what would drain the battery without it.

DEMO PROGRAM

The “Show what I can do” button plays 30 seconds of a demo in the spirit of the demoscene, in real time: a flyover of Atlas terrain, a hundred windows with live video in Flip Through, thousands of snowflakes and Dijana. At the end, one sentence: “This ran while the processor was almost idle.”

SHOP WINDOW

A mode for shops and fairs that loops the demo and invites a passer-by to move a window or throw a document into the Bin.

Rules

Beautiful, but not a burden

LEVELS

Calm: no motion and no glass, for work and weaker eyes. Lively (default): glass, tiles, short animations and window physics. Lavish: everything, plus seasons, the 3D overview and a live background.

SPEED

No animation lasts longer than 150 ms and each one is interrupted by the next click. Work never waits for an effect.

POWER

On battery and in Efficient mode the system lowers the level by one by itself. Estimate for Neo 2000: Lively takes under 3% of the GPU, Lavish under 10%, and the main processor almost nothing.

ACCESSIBILITY

“Less motion” and “No transparency” apply across the whole system and in all apps from Neo Kit.

ORPHEUS

Sound as part of the experience

Good sound tells you what happened before you look: it arrived, it worked, it failed, it's done. Neo sounds are handled by the Orpheus chip, with latency under 5 ms, so the sound arrives exactly with the motion on screen. The rule is the same as for beauty: useful, short and quiet, and every sound also has a caption.

Sound comes from where the thing is on screen, and music lowers itself



Example sound themes made by synthesis, CC0, free with the system

Theme	Starting up	Open	Notification	Error	Bin	Done
Crystal glass bells, for Glass	▶	▶	▶	▶	▶	▶
Wood marimba, warm and quiet	▶	▶	▶	▶	▶	▶
Chip 8-bit, like the home computers of the 80s	▶	▶	▶	▶	▶	▶

Meaning

Every sound says something

LANGUAGE

Success goes up, an error goes down and short, a notification is two tones, and a finished long job is a small chord. The same rule applies in every theme, so the meaning is learned once.

MEASURE

Frequent actions (click, open) last under 150 ms and are barely audible. Longer sounds only for rare moments: power-on, a finished backup, an error.

HARMONIOUS

All sounds of one theme are in the same key, so even when two overlap they sound like a chord, not noise.

Space

Sound from where the event happens

PANORAMA

A notification from a window on the right is heard on the right, and from another monitor from that side. On headphones Orpheus does it in 3D.

DUCKING

When a message arrives or the Neo voice speaks, music and films dip for a moment, then come back.

RHYTHM

System sounds don't pile up: ten notifications in a row are heard as one, with a number on screen.

Physics

Sound follows Newton

MOTION

A thrown window rustles quietly, louder the faster it is. A crumpled document rustles in flight and lands in the Bin with a dull thud, and a bigger file sounds deeper.

PROGRESS

Optionally, a long copy has a quiet tone that rises as the job progresses, so you know how far it has got without looking.

GAMES

Pinball, cards and pieces get their sound from the same physics, not from recorded samples.

Dijana

A resident you hear, but who doesn't talk

SOUNDS

Sniffing while searching, a yawn when waking up, a short “woof” when the backup is done and tail thumping. No words and no tiresome sentences.

HOT, COLD

During a search the sniffing gets faster and higher as it nears the right result, like the game “hot and cold”.

Ambience

A soundscape if you want it

DAY

A quiet background that follows Dijana's Day and the real weather outside: rain when it rains, birds in the morning, crickets in the evening. Off until you turn it on.

FOCUS

Noise for concentration, a café in the distance or a library. A work and break timer with a soft bell.

Accessibility

Sound that guides

SCREEN READER

Short tones for a window edge, the end of a list and a button that can't be pressed, so a blind person hears the shape of the screen.

CHARTS

Table and Atlas can “play” a chart or terrain height, with a tone that rises and falls.

CAPTION

Every sound has a text and visual counterpart, for those who can't hear.

Silence

Silence is a feature

BUTTON

The Quiet button on the case and keyboard turns off all system sounds, and the music stays.

NIGHT

After 10 pm sounds are automatically quieter and softer, and during a call in Chat the system stays silent.

Time of day

Sound that follows the day

IDEA

Optionally, the theme changes colour through the day: wood and brighter tones in the morning, softer and lower glass in the evening. The meaning of the sounds stays the same.

STARTING UP

The power-on sound changes slightly with the seasons and holidays, and your own one from the Orpheus synthesizer stays as it is.

Retro

Real sound chips in Orpheus

CORES

Orpheus can load FPGA cores of classic home computer sound chips, so the Chip theme and retro games sound like they used to, with real noise and filter.

FOR AUTHORS

A theme can also be written as a small music file for such a chip, a few kilobytes in size.

Sound themes

Themes in the Treasury, like Residents and alongside them

WHAT A THEME IS

A pack of up to 40 sounds for system events, plus power-on and shutdown sounds and an optional ambience. Free with the system: Neo Basic, Crystal, Wood and Chip.

BUYING

In the Treasury, once and forever, without a subscription. The author gets most of the revenue, so local composers and sound designers get a market. Community themes come free under a free licence.

WITH A RESIDENT

Every resident can bring its own theme, and they are also sold separately. The dancing snake resident from “Ide zmija” gets a theme with parts of the song for power-on and a finished job, but only with the official licence of DJ Nick DeeTronic.

MIXING

Each event can come from a different theme: power-on from Chip, the Bin from Wood. An app can have its own theme, for example Arcade.

OPEN FORMAT

A theme is Opus or FLAC files and one text list of events. It is made in Wave or the Orpheus synthesizer and shared as a single neopak file.

CHECKING

The Treasury checks duration and loudness (all themes equally loud) and that an error is clearly distinguishable from success. Themes may not contain ads or speech that sells something.

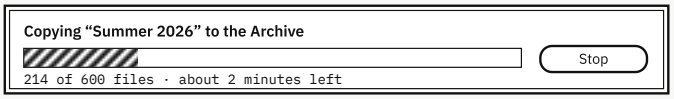
EVEN EASIER

Hands, voice and little things that take away worry

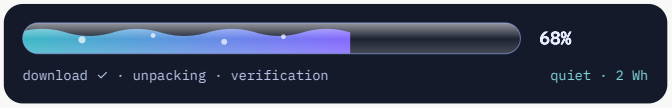
Every input does what it is best at: the keyboard writes, touch arranges, the pen draws, and the voice commands. Along with that come cheap hardware extras and system parts that are the same in every app. We took the ideas for shared parts (progress bar, pie, collapsible toolbars, draggable tabs, a Shelf you drag files onto) from the Enhancer package for AmigaOS 4, but with our own names and look.

Progress bar: beautiful, interesting and honest

Retro clear and honest, like on the old Mac



Future Shock liquid in glass, Newton makes it ripple

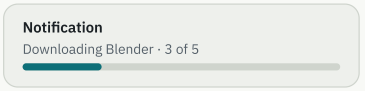


With Dijana a long job, a short story



Dijana carries the files to the bone at the end. When she gets there she wags her tail and you hear “done”.

The same progress everywhere





icon on the Shelf fills like a pie

- **Honest** measured by the Swarm, not guessed; no jump from 99%
- **Says what it's doing** stages and file, no endless spinning
- **Always “Stop”** and “Continue later” after a restart
- **One part of the system** he same in every app and both styles
- **Sound** a quiet rising tone, “done” at the end
- **Calm** no motion at the Calm level and with reduced motion

Hands

The keyboard writes, touch arranges

KEYBOARD

A real mechanical keyboard is the main input for text. Glass has no tactile feel, so you type slower and with more mistakes.

TOUCH

For maps, pictures, piles, Tiles and kids. When you touch the screen, Easy mode enlarges the targets by itself. Touch isn't the main input on an upright monitor, because the arm tires quickly.

PEN

The most precise for drawing and signatures. The Neo monitor for the 4000 (option) folds down into a drawing table.

Voice

A “talk” button instead of constant listening

HOW

You hold the button like on a walkie-talkie and say a command or dictate a rough draft. Corrections are made with the keyboard.

WHERE IT'S GOOD

Commands, dictation, accessibility and hands busy in the workshop or kitchen.

PRIVATE

Everything is recognized locally, and an LED shows when the microphone is listening.

Feel

Feeling without glass

TRACKPAD

Vibration is driven by Newton: you feel it when a window “clicks” against an edge or when you drag a file into a folder.

DIAL

On the case and keyboard: volume, zoom, time in Montage and brush size in Brush.

GAMEPAD

An equal input for the couch, for the whole system and not just for games.

Tap

NFC cards as physical shortcuts

IDEA

You tap a card and the system does one thing: opens a child's profile, starts a game or begins the “Taxes” task. The reader costs about €2.

FOR WHOM

Kids, older people and everyone who likes objects, in the spirit of the cartridge.

MAKING THEM

You make a card yourself in Helpers, and the Treasury has ready-made ones with pictures.

Guest port

An unknown USB device can't do harm

HOW

An unknown stick opens in an isolated room and shows only files.

FAKE KEYBOARD

If a device presents itself as a keyboard, you first type a code from the screen. That way BadUSB doesn't get through.

ONE CABLE

USB4 to the monitor carries picture, power and peripherals. One button switches the keyboard and mouse between Neo and Satellite.

Call your people

Help from someone you trust

HOW

One click and your child, a friend or your club can see your screen. At first they only watch, and to take control they ask for your “yes”.

SAFE

End-to-end encrypted, without intermediaries, and it switches itself off after 30 minutes. Folders labelled Personal are blurred.

Record what I do

Automation without programming

HOW

You turn on recording, do the job once and get a Helper or a Lux script.

LEARNS

The script is readable, so you see how it works and can change it. That is the road from user to programmer.

Progress

A bar that's nice to watch

HONEST

Swarm knows how much work is left, so there is no jump from 99% and no endless spinning. Before a long job you see an estimate (“about 4 minutes, quiet, 2 Wh”).

BEAUTIFUL

Retro stripes, liquid in glass or Dijana carrying files. The same system part in every app, notification and icon.

ALWAYS

A “Stop” button and “Continue later”, which survives even a restart.

Parts

Shared parts for all apps

WHAT

Progress, a pie for disk usage, media (play, stop, seek), a skinnable clock and one Viewer for all formats.

TOOLBARS

When a window is narrow, buttons that don't fit move into a menu behind an arrow, and the most important ones stay visible.

TABS

They are dragged to change the order or pulled out into a new window. Each has a close button and a description on hover.

WHY

The developer doesn't build their own, and the user sees the same thing everywhere. When a part is fixed, it is fixed in every app.

Shelf and notifications

Drag onto the icon and you're done

DRAG TO OPEN

Drop files onto an app's icon on the Shelf and they open in it, even several files at once.

DRAG TO COPY

Drop them onto a folder on the Shelf: they are copied, it only asks when the name already exists, and the folder opens with the new file selected.

EDITING

When the Shelf is unlocked it has a visible frame, so you know it is being changed. Multiple shelves and sub-shelves, each can be collapsed into a strip.

NOTIFICATIONS

Progress right in the notification, and sound set per app: as the system, its own or silence.

RESTART

After an installation the system says clearly whether a restart is needed and why. Usually it isn't.

Eternal format

A document that will still open in 30 years

HOW

It is saved in an open format, along with PDF/A and plain text.

WITHOUT NEO

It opens on any other computer too, without our app.

First week

Seven small tasks with Dijana

WHAT

A picture, a letter, music, a backup, the Board, a Helper and a short typing game.

RULE

Every task can be skipped and never comes back on its own.

New generations

Used to touch, they learn the rest too

WHAT

Phone habits work (swiping, zoom, sharing), and Neo teaches typing, files and shortcuts along the way.

SPIRIT

Teaches, doesn't dumb down: nothing is hidden, the order is just gentle.

Part prices are estimates. The names AmigaOS, Enhancer and BadUSB only describe where an idea came from and are not used in the product.

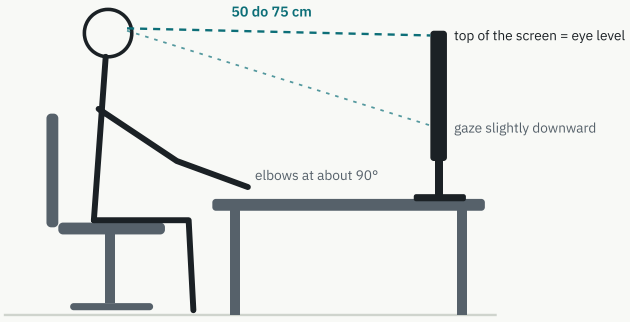
A COMPUTER FOR PEOPLE

Screen, eyes and hands: a smaller price for your health

Neo should be easy to learn, efficient without distractions, focused on knowledge and tasks and gentle on the body. We don't promise this, we measure it and publish the results. For the eyes, the type of panel matters less than five things: no flicker, a matte screen, light matched to the room, sharp text and a good posture.

Good posture and Two-Face, the screen with two faces

A posture that protects your eyes and back



Light from the side, not behind the screen. Every 20 minutes, look into the distance for 20 seconds.

Two-Face: one screen, two faces (future concept)



Paper: light from the room — **Colour: IPS for graphics and games**

Today	e-ink as a second screen or Paper mode on IPS
Tomorrow	transflective LCD: light off = paper, light on = colour
Precedent	the OLPC XO-1 screen (2007) worked exactly like that, in two modes
Neo	Panorama picks the face per app: Quill → Paper, game → Colour

Neo display

27" 4K IPS, matte, 120 Hz

WHY IPS

Text scrolls smoothly, colours and viewing angles are good, and there is no burn-in of the Shelf and menus as with OLED.

NO FLICKER

Dimming with direct current, without PWM, even at the lowest brightness.

OTHERS

About 160 ppi with 2× scaling, VRR, light sensor, colour calibration, a stand that raises and rotates.

Paper

For long texts and reading

PAPER MODE

On every screen: shades of grey, high contrast, no motion and no colourful icons.

E-INK DISPLAY

An optional second screen for Leaf and Quill. It has no backlight, so it is the easiest on the eyes, but it is slow for films and games.

Two-Face

A hybrid screen: paper for text, colour for games

IDEA

When the backlight is off, the screen reads like paper in the room's light. When it is on, it is a colour IPS.

THE WAY THERE

Transflective LCD. The screen of the OLPC XO-1 laptop from 2007 worked exactly like that, and reflective LCDs exist today in reading tablets.

NEO

Panorama chooses the face per app by itself: Quill and Retro 2D on Paper, Brush and games in Colour. A goal for Gen 2 or 3, with a panel partner.

Hands

Peripherals for 10 years

KEYBOARD

Mechanical, replaceable keycaps and cable, with a low-profile and a split variant.

MOUSE AND PEN

A symmetrical mouse and a vertical variant. A battery-free pen (EMR) with a replaceable tip.

PARTS

Switches, keycaps, wheels and tips in the Treasury and at service centres.

Focus

One task, no distractions

HOW

With “What are you doing today?” only the windows and notifications for that task remain. Other notifications arrive in a bundle, once an hour.

NO BAIT

No endless scrolling, red counters or autoplay in system apps.

Health

Gentle on the body

LIGHT

Panorama adjusts brightness and colour warmth to the room and the time of day.

BREAKS

Dijana occasionally asks for a walk: 20 seconds of looking into the distance every 20 minutes and getting up every hour. It can be turned off.

SETUP

The first week shows how to set the height of the screen and chair, with a picture.

Proof

We measure and publish

WHAT

Time to complete a task, number of errors, how fast a beginner learns, number of interruptions per hour and eye strain by a standard questionnaire.

HOW

An independent lab or university, before launch and every year, compared with an ordinary computer for the same job.

PUBLIC

The results and the method are public, even when they aren't good.

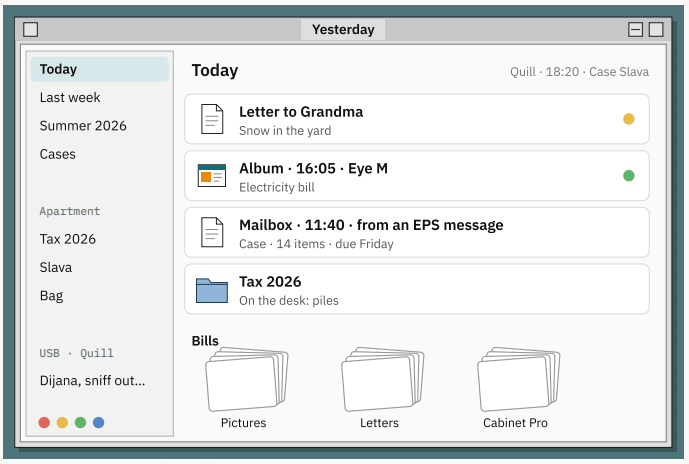
Measures and goals are concept estimates. Two-Face is a future concept and depends on panel manufacturers.

CABINET

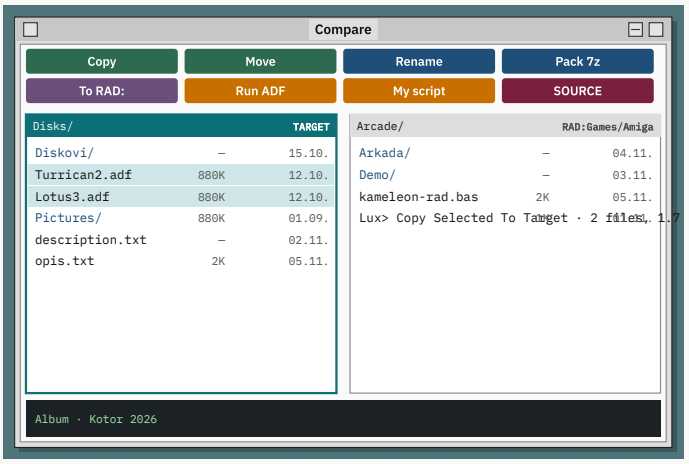
Files without hassle, and the whole toolkit for experts

Two faces of the same Cabinet. In Easy mode there are no files, extensions or saving, only documents, as on the Lisa: you work, and Neo remembers. In Pro mode Cabinet becomes a custom tool in the spirit of the last Directory Opus on the Amiga, with listers, button banks and modules.

Easy: documents by time, piles and Cases



Pro: listers, button banks and modules



No saving

No “Save” and no file names

REMEMBERING

Every change is remembered immediately, in RAD: and then on the SSD, and versions are made automatically. The “Save” button doesn't exist, as on the Lisa.

NAME

A document gets its name from the first line or title, and a photo from the place and date. You rename it when you want, you never have to.

NOVO

A new document is torn off a Template, like a sheet from a pad. You don't choose a program, you choose a kind of paper.

One place

A file lives in only one place

LINKS

“Also put in...” and “Send” make a link, not a copy. A change is seen everywhere, so there is no “contract_final_2_real”.

DUPLICATES

When a file arrives that you already have, Neo says “You already have this in Work” and doesn't make a second copy.

BAG

An external disk and a cartridge are a Bag: what you put in it, you carry with you, and Neo remembers what went where.

Views

By time, by person, by job

TIME

Today, Yesterday, Last week, Summer 2026. People remember when they did something better than where they put it.

PERSON

Everything from Grandma or from the accountant in one place: messages, files and pictures.

LIVE FOLDERS

A folder that is actually a question (“bills this year”) and fills itself.

On the desk

Piles, colours and Cases

PILES

Files on the desk stack into piles by type or day, like papers. Swiping a finger flips through a pile.

COLOURS

Seven colour labels with names you choose yourself, as on the old Mac.

CASE

A project folder that holds files, messages, notes, deadlines and people together: “Flat”, “Slava (family feast)”, “Taxes 2026”. Close the Case and everything is put away.

Simple

Little things that take away worry

EXTENSIONS

Hidden in Easy mode. A file is recognized by its icon and preview, and dangerous files never disguise themselves as documents.

SPRING

A folder opens when you hold a dragged file over it, so you go deeper without letting go.

NOTHING GETS LOST

Moving and deleting can be undone, the Bin keeps things for 30 days, and Dijana sniffs out where a file went.

Cabinet Pro

In the spirit of Directory Opus on the Amiga

LISTERS

As many listers as you want, each with its own view, and any two become source and destination.

BUTTON BANKS

Banks you build yourself: a button runs a command, a Lux script or a series of steps, with left and right click.

FILE TYPES

Every type (picture, floppy, archive, code) has its own actions for double-click, drag and menu.

Modules

A Cabinet that grows

BUILT IN

FTP and SFTP as an ordinary lister, comparing and syncing two folders, batch renaming with a preview before running, finding duplicates, viewers for all formats and working with archives as with folders.

LUX

Every action has a Lux command, like ARexx on the Amiga, so Cabinet can be controlled from other programs and by voice.

SHARING

Banks, scripts and modules are shared via the club Board, signed and sandboxed until reviewed.

Two faces

Easy and Pro on the same files

SWITCHING

One switch, without losing settings. In Pro mode you see the real names, paths (Foundation is System:), extensions and permissions.

SAME DATA

In Pro mode a Case is an ordinary folder with labels, and a pile is a view, so nothing is ever locked into a format only Easy understands.

Lisa and beyond

More ideas from that school

CARD FILE

Contacts, notes and recipes in one system card file that every app reads, as on the Apple Newton of the nineties.

SEND AS A PACKAGE

“Pack for mail” shrinks pictures and compresses files to the mail limit by itself.

WHERE IT CAME FROM

Every file remembers where it came from (message, website, camera) and shows it with one click.

GUEST ROOMS

Runs almost everything, with one double-click

The user doesn't choose an emulator, a compatibility layer or a virtual machine. Double-click a program or installer, and Neo itself chooses the room where it runs best, just like Arcade for games. Each guest lives in its own Guest room and sees only what you give it a Key for.

What you run	The room Neo chooses	How it works	Honest
Windows programs and games	Bridge + Wine/Proton	x86 code is translated to ARM on the fly, DirectX goes via Vulkan to VSA-6000	Most programs and games work, and the “Does it work?” database tells you in advance
Windows that doesn't work that way	Full Windows in a box	On Neo 2000 and 4000 with the Satellite x86 module at full speed, otherwise in emulation	The user buys the Windows licence
Linux programs	Apps domain	deb, Flatpak and AppImage, the windows behave like Neo windows	Almost everything
Android apps	Android room	Open Android in a separate room, stores of your choice, microG instead of Google services if you want	Apps that require Google device verification (some banks) don't work
DOS, Amiga, C64, consoles	Chameleon and Arcade	FPGA core if there is one, otherwise an emulator	The user brings the games and ROMs
Web apps	Horizon	Installed like ordinary programs, with an icon on the shelf	Everything that works in a modern browser
Games with kernel-level anti-cheat	None	That anti-cheat demands full control over the system, which clashes with Fortress	Doesn't work, unless the publisher allows running under Proton
macOS and iOS programs	None	The licence ties them to Apple hardware	Doesn't work
Streaming films at the highest resolution	Horizon	Requires certified content protection in hardware (Safe could do it, with a contract)	Lower resolution until certification

One click

The user doesn't choose the room

ORDER

Neo checks the “Does it work?” database, then tries Bridge and Wine, then the Android room or a full box. If nothing works, it says why, in plain language.

PRO

In Pro mode the room is chosen by hand, with settings like in the Proton tools.

Does it work?

A shared database of experiences

HOW

After the first run Neo asks “How did it go?”, and the rating and settings go into the database on the Board, like the shared compatibility boards for games on Linux.

PRIVACY

Only the program name, version and rating are sent, and only if you agree.

Security

The guest doesn't see the house

ROOMS

Each guest has its own part of the disk and sees your folders only when you give it a Key. A virus in the Windows room stays in the room, and Chronos takes it back to yesterday.

RULES

Lock labels apply to guests too: Personal and Work files don't enter the room without asking.

BORROWED AND IMPROVED

The best of macOS X, Darwin and Linux

We already have search, Chronos (going back in time), Flip Through, the app as a single file and Keykeeper. These are the next good ideas from the Unix world, under our own names and adapted to Easy and Pro mode.

macOS X

Looking and acting without opening

SPACE BAR

Select a file and press the space bar: a quick preview of a picture, PDF, film, sound or spreadsheet, without opening a program.

SERVICES

The Services menu offers actions on the selected text or picture in every app: translate, read aloud, send to Quill, make a task, sniff out.

RECIPES

A series of steps without programming (“shrink the pictures and send them to Grandma”), assembled by dragging blocks and run through Lux.

macOS X

Little things for the hands

TITLE ICON

The small document icon in the window title can be dragged into a message, onto the Board or into another folder, as if you were carrying the whole document.

HOT CORNERS

Mouse into a corner: Flip Through, the desk, Nap or the screen saver, as you choose.

DICTIONARY

A long press on a word opens the dictionary and Wikipedia, without the internet.

macOS X

Several Neo devices as one

CONTINUE

Start a letter on the Neo 500, continue on the Neo 2000 with one click. Clippings work on both.

HAND TO A NEIGHBOUR

A file is sent to a nearby Neo or phone through the Neighbourhood, without a cable and without the internet.

ONE MOUSE

The same mouse and keyboard move from one Neo to another across the screen edge, and a tablet becomes a side screen.

MOVING HOUSE

A wizard transfers files, mail, pictures and settings from the old computer, from Windows and Mac too.

DISK MODE

A Neo connected by cable becomes a disk for another computer, for rescuing data when the system won't start.

Darwin

Devices without setup

NEIGHBOURHOOD

Printers, scanners, speakers and shared folders on the network show up by themselves, as with Bonjour. No hunting for IP addresses.

PRINTING

Driverless printing over the open IPP standard, so it works with almost every new printer.

Darwin

A smart file system

CLONES

A copy of a file is instant and takes no space until it changes. Copying a 20 GB film takes a fraction of a second.

METADATA

Labels, colours, origin and notes travel with the file, on the Bag and through the Board.

DUAL PACKAGES

One neopak carries code for ARM and for Satellite x86, so the system picks the faster one.

Darwin

Order under the hood

HELPERS

One manager for all services, which start only when needed, like launchd.

MAGNIFIER

In Pro mode, live tracing of what the system and programs are doing: files, network, power. No restart and no special tools.

LOG

One system log, with personal data hidden by default.

Linux

Habits that save time

MOUSE

Middle click pastes the selected text, and Alt plus drag moves a window wherever you grab it. Switchable in Pro mode.

SHELL

Every command has short help with examples, like tldr, before the long manual.

SCREEN

Night light and power profiles (Quiet, Balanced, Turbo) from the menu.

Linux

Security without questions

PORTALS

A program doesn't see the disk. When it asks for a file, a system dialog opens, and the program gets only the file you choose. It's a Key you don't need to understand.

FIRMWARE

The firmware of all devices, external ones too, is updated through Renewal with a signature, like fwupd on Linux.

Linux

Neo in your pocket

LIVE SYSTEM

The whole Neo OS on a cartridge or USB disk, starts on another computer without installation, with your Workdesk.

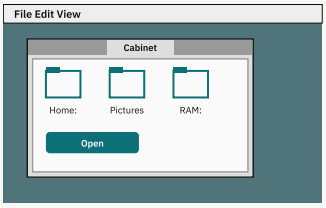
TRY IT

A school or shop can let people try Neo on their own computer before buying.

LOOK AND VOICE

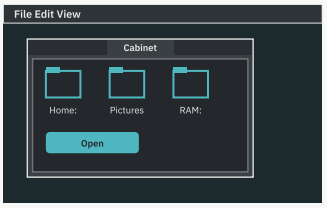
Themes, voice and retrofuturism

Colours change with one shortcut or from the menu, instantly and in all apps, and work in both styles (Retro and Future Shock). It can also be automatic: Slate after sunset.



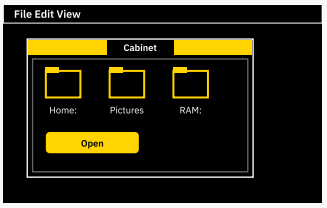
Silver

Light, the default



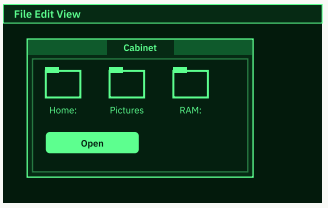
Slate

Dark, for night



Contrast

High contrast, for weaker eyesight



Phosphor

Retro, green monitor

Voice

Dictation and commands

DICTATION

Into any text field, in Serbian, in Latin or Cyrillic, with punctuation. The model runs locally on Neuron.

COMMANDS

“Open Cabinet and show yesterday's pictures.” Voice is translated into Lux commands, so it works with every app.

PRIVACY

The wake word is off until you turn it on yourself. Nothing goes to the internet.

Microphone

Sound worth hearing

INCLUDED

A microphone array in the Eye camera (2 to 4), with beamforming toward the speaker.

NEO MICROPHONE

A desktop condenser microphone with USB-C and XLR outputs, a physical mute button and an LED ring.

ORPHEUS

A quality preamp and noise removal in the DSP, without loading the processor.

Retrofuturism

Extras with soul

SCREEN

CRT mode in Panorama: scanlines, a slight glow and curvature for the whole screen or just one window.

TELETEXT

News and the weather forecast in teletext style, with page numbers.

SOUND

A power-on sound from the analogue synthesizer in Orpheus and an optional robotic reading voice, like the speech chips of the eighties.

OTHERS

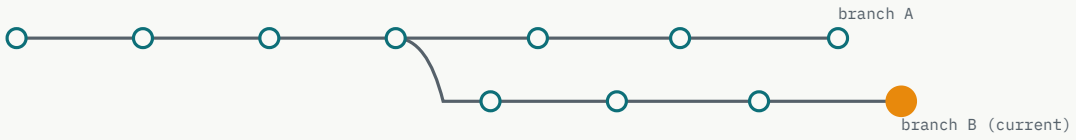
Neo Meditation: an error screen that explains what happened. A screen saver with demoscene effects and a BBS client for the community.

UNDO AND TRANSFER

Unlimited undo and datatypes

Undo works on three levels: an action in a document, a file version and the state of the whole system. Data transfer between programs goes through system datatypes, in the spirit of the Amiga datatypes.

A history tree in the document: nothing is lost even when you go back and continue down another path



1 · Action
Unlimited Ctrl+Z, saved with the file

2 · File
every save is a version (copy-on-write)

3 · System
Chronos: the whole computer as it was yesterday

Undo

Without limits and without fear

DOCUMENT

Every change is recorded as a step. A time slider brings the document back to any moment, and the history stays even when you close the file.

SISTEM

Moving, renaming and deleting in Cabinet, as well as changes to settings, can also be undone.

LIMIT

Sent mail can't be brought back. That's why Pigeon waits 30 seconds before sending, and such actions are clearly marked.

Datatypes

Every program understands every format

HOW

Formats are system add-ons. When you install a datatype for WebP or DjVu, all programs understand it at once, old and new.

COPY AND PASTE

A clipping carries several forms at once (text, table, picture, contact), and the receiving program takes the one that suits it.

EVERYWHERE

It also works with Windows windows through Clasp and with retro cores: text is “typed” into the 68030 system, and a picture is converted into its format.

More help

Little things that save your nerves

COLLECTION

Clipboard history with search. Passwords disappear from it after a minute, and Lock labels apply to clippings too.

LIVE LINK

A table from Abacus pasted into Quill stays linked and refreshes, like Publish and Subscribe on the old Macs.

NO LOSS

Autosave everywhere and all windows restored after a restart. “Save” only means “give this version a name”.

SEARCH

One search for files, messages, settings and commands, in plain words, locally.

BEST OF THE PAST

Good old ideas, done the modern way

Ideas that were ahead of their time and disappeared with the systems that had them.

Idea	From	In Neo OS
HyperCard	Apple, 1987	Neo Cards: apps made of cards and buttons without programming, for recipes, collections and school projects.
Services from the menu	NeXTSTEP	Select text or a picture in any program and choose what other programs can do with it: translate, send, convert.
Smart folders	BeOS	Files carry attributes like a database, and a folder can be a permanent search (“all invoices from 2026”).
Screens that slide down	Amiga	Pull the desktop down and underneath you see a retro core or the Windows layer from Panorama.
Window shade and spring-loaded folders	Mac OS 8/9	Double-click the title to roll the window up into a bar. A folder opens by itself when you hold a dragged file over it.
RAD: disk	Amiga	A working disk in RAM that survives a restart. The contents remain after updates and program crashes, and back then you could even boot from it.
Drag to save	RISC OS	The icon from the save window is dragged into a folder. No wandering through a dialog.
Look underneath	Smalltalk	Right-clicking any element shows which Lux command it runs, so every action teaches you how to automate it.
Command palette	modern editors	One shortcut opens a field where you type the name of an action, program or file, and it works in every app.
Focus mode	modern phones	One switch silences notifications and hides everything except the work you are doing.
Accessibility from day one	modern OSes	Screen reader, magnification, captions for every sound and voice control, turned on at first power-on if needed.

MODERN NEO

Six principles and what they change

Each principle gets a part in the OS and a part in hardware. The hardware part is there so software can't bypass it.

Flow

It doesn't matter where a program runs

OS

The Dispatcher chooses where to run by itself: native ARM, x86 translation, Satellite, an FPGA core or another Neo in the house. The user only sees an icon and launches it.

HARDWARE

Coherent memory (CXL) between the ARM, the FPGA and Satellite, so a program moves without copying data.

Renewal

One update for everything

OS

One signed system image: the kernel, all drivers, Chameleon cores, Satellite firmware and VSA-6000 microcode. It is written to the other partition while you work, and going back is one button. Drivers as a separate item don't exist.

HARDWARE

Dual Neo ROM (A/B), so even an interrupted update doesn't leave the machine unable to boot. Service mode via USB-B.

Treasury

Open source

OS

Every app in the Treasury has a Source code button and a reproducible build: the system checks that the installed program matches the published code. Neo Kit and Lux under free licences.

HARDWARE

Open schematics for the cartridge, AGP-X and the Satellite connector. FPGA cores published as source code.

Fortress

A safe, fast and stable kernel

OS

A formally verified microkernel (seL4 class) at the bottom. Exec-N and the system services in Rust. Linux remains only as an isolated driver domain: when a driver crashes, the domain restarts, not the system.

HARDWARE

ARM memory tagging (MTE), an IOMMU for every device, encrypted RAM. Satellite and PCIe cards see only the memory assigned to them.

Lock + Safe

Lock against data leaks

OS

Every file carries a label: Personal, Work or Public. Labelled data doesn't leave the machine without confirmation. A per-app firewall shows who sent what.

HARDWARE

The Safe chip keeps the keys and encrypts the disks. Physical switches for the camera, microphone and network cut the power, and software cannot turn off the LED next to them.

Eye · Montage · Broadcast

Camera, video and streaming

OS

Panorama layers serve as scenes for streaming. Neuron does background removal, framing and subtitles in Serbian. Montage works on proxy footage, without waiting for a render.

HARDWARE

An Eye camera included with every model (S, M, Pro), with a mechanical shutter, HDMI input for capturing another machine, AV1 encoder in Panorama, Orpheus microphone preamp.

NEO ROM 2.0

The OS in ROM and the Neo cartridge

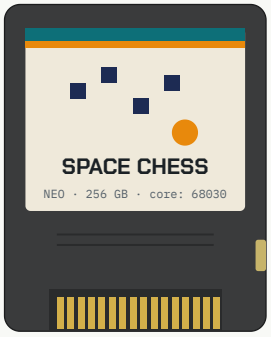
The system and the basic apps live in a soldered, read-only flash chip, like Kickstart back then. The SSD keeps only user data. The cartridge is a fast physical medium for games, apps and FPGA cores, and digital purchase through the Treasury remains an equal option. Speeds and capacities are concept estimates.

Memory layers and boot

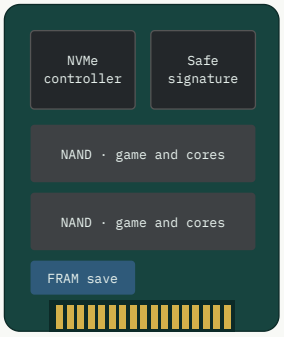
NOR ROM 64 MB	Kernel, Exec-N, basic GUI executes directly from the chip · works even without a disk
pSLC ROM 2 × 32 GB · A/B	All of Neo OS + Quartet, Horizon, Montage read-only · written only by Renewal
Cartridge 64 GB – 1 TB	Games, apps, FPGA cores works instantly, without installation
SSD mirror 2 × M.2	Only user data and additional apps one disk fails, work goes on
Guardian home server	Backup over the network encrypted with the Safe key

Boot: NOR → Chronos snapshot from pSLC → desktop, about 1–2 s

Neo cartridge



outside · write-protect on the side



inside · PCIe 4.0 ×2

About 3 GB/s read · the game starts directly from the cartridge

	Neo cartridge	UHD Blu-ray / M-DISC	Digital (Treasury)
Capacity	64 GB to 1 TB	50 to 100 GB	As much as there is room on the SSD
Reading	About 3 GB/s	About 100 MB/s	SSD speed
No installation	Yes	No, copied to the SSD	No, downloaded
Durability	Good, pSLC	M-DISC: archival, for decades	As long as you keep a copy
Ownership	Can be lent and sold	Can be lent and sold	DRM-free package, stored locally
Cost to the publisher	Highest	Low	Lowest
Best for	Games, cores, collector's editions	Films and archives	Everything else

NEO PERCEPTION

Enough for the eye, no more

Hardware that doesn't waste power on differences people can't see or hear. Efficient mode is the default and gives smaller game downloads, smaller files, more frames per second and lower power use. Pro mode remains for professional work and is turned on per app. The hardware supports both modes.

	Efficient (default)	Pro	Hardware
Screen	Renders at a lower resolution, and Neuron upscales to 4K. 8 bits with dithering.	Native resolution, 10-bit HDR, calibrated colours	Panorama scaler, Neuron
Textures in games	ASTC compression, 4 to 8 times less memory	Uncompressed or lightly compressed textures	VSA-6000 decoder
Shading	Variable rate: detail where the eye is looking	Every pixel at full quality	VSA-6000
Sound	Opus, the same as the original to the ear	Lossless FLAC, 24-bit	Orpheus decoder

	Efficient (default)	Pro	Hardware
Pictures	JPEG XL / AVIF, 5 to 10 times smaller	RAW and lossless JPEG XL	Courier codecs
Video	AV1	ProRes class for editing	Panorama encoder

Games

Two content packs

TREASURY

A game is downloaded with the Efficient pack, roughly half the size. The Pro texture and sound pack is an add-on for those who want it.

CARTRIDGE

More games or richer content fit on the same cartridge.

Rule

The original is never lost

KEEPING

Lossy compression is one-way, so the system always keeps the original when you have it. The efficient copy is for working and sharing.

LABEL

Every file clearly shows whether it is Efficient or Pro.

Energy

Longer on battery and sun

CONSUMPTION

Efficient mode uses noticeably less power, so the Neo 500 runs longer on battery and the Sun station lasts longer.

AUTOMATIC

When power comes from the battery, the system switches to Efficient mode by itself, except in apps marked as Pro.

COURIER

Compression everywhere, the system from RAM

The spirit of the old crunchers, but at the level of the whole system. Courier decompresses in hardware faster than flash can read, so compressed data actually speeds the computer up.

ROM image

A compressed system

HOW

Neo OS is one EROFS image compressed with LZ4/zstd, like the system partitions on Android.

GAIN

About twice as much system and apps in the same pSLC chip.

WHY NOT PER PROGRAM

Compressing each EXE separately (like UPX) breaks memory sharing and signatures, so it is done at the file system level.

Neopak

An app is one file

PACKAGE

Every app is one compressed, signed file that mounts instantly, without installation and scattered files.

DELETING

You delete one file and nothing is left behind.

ARCHIVES

Cabinet opens ZIP, 7z and RAR as ordinary folders. New archives are made in the open 7z and zstd formats, because RAR is a closed format.

Like MorphOS

The whole system at once

STARTING UP

Courier loads and decompresses the system core (2 to 3 GB) into RAM in one go, so everything runs from memory.

OPTION

“Whole system in RAM” on Neo 2000 and 4000, with your most used apps preloaded.

MEMORY

Compressed swap in RAM (zram) gives about 50% more space before the SSD is touched.

Like the Amiga

A RAD: disk that survives a restart

WHAT IT IS

The RAD: volume (working disk) in RAM, as fast as memory. Unlike RAM:, it survives a warm restart, a program crash, an update through Renewal and a change of FPGA core.

HARDWARE

The memory controller keeps the RAD: zone in self-refresh while the rest of the system resets, and Neo ROM doesn't erase it at boot. The zone is encrypted with a key from the Safe chip, and every block has a checksum, so a damaged block is recognized.

SIZE

Grows and shrinks as needed, compressed via Courier. Proposed limit: 4 GB on Neo 500, 16 GB on Neo 2000 and 32 GB on Neo 4000.

SHUTDOWN

A cold shutdown erases RAD:, unless “Save RAD: at shutdown” is on. Then Courier compresses it to the SSD in a couple of seconds and restores it at the next power-on. The Neo 500 on battery keeps it even when the power goes out.

FOR WHAT

Temporary projects and compiling programs, floppy and disk images for retro cores (Chameleon can boot directly from RAD:), the Horizon and Arcade cache and resuming work after a restart.

CLEARING

A long press on the Chronos button at power-on or the “Empty RAD:” command. Lock labels apply here too.

Standard format

One compression format for the whole system

ALGORITHMS

Open zstd for almost everything and LZ4 when speed matters most. Courier does them in hardware in both directions, compression and decompression. We don't make our own algorithm, so that Neo files can be opened on any other computer.

WHERE IT'S USED

File system, ROM image, neopak apps, Chronos snapshots, RAM (zram), network transfer, backups and cartridges.

FOR DEVELOPERS

One call in Neo Kit, “Compress”. The system chooses hardware or software by itself, so future apps get the acceleration automatically too.

FUTURE

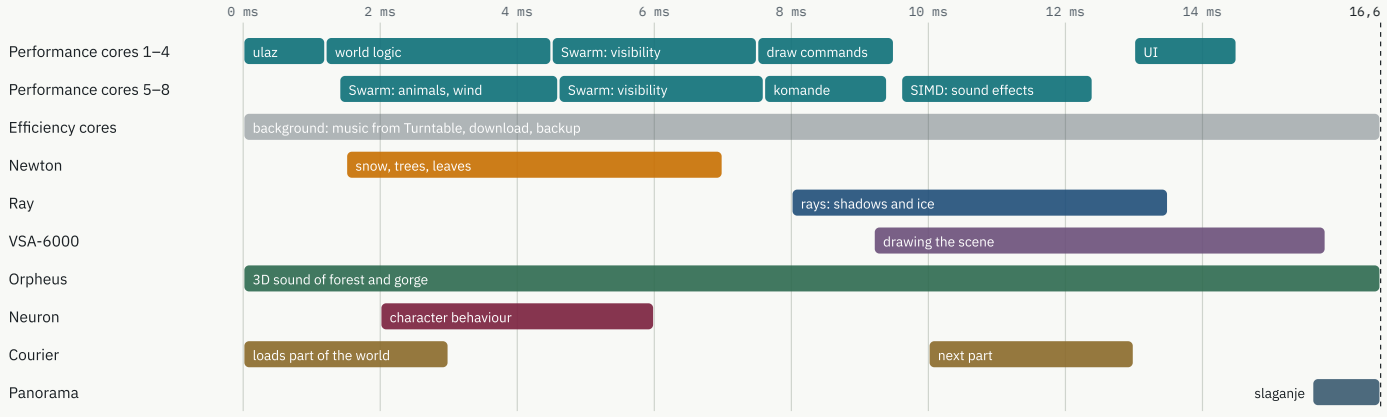
Every compressed block carries an algorithm and version tag. A new algorithm is added through Renewal, and old files stay readable forever.

SWARM

Processor, threads and multitasking: everything is a series of small jobs

Under the hood Neo splits everything into small jobs that last a fraction of a millisecond. One job queue fills all cores and all chips at once: a short job stays on the processor and runs with SIMD instructions (NEON and SVE2, the modern successors of MMX), and a long data stream goes to the chip made for it. It is the spirit of BeOS, where everything was threaded, but with today's job scheduling instead of manual thread management.

One frame of the game “Storm over Zlatibor” (16.6 ms at 60 frames per second): who does what



Example and estimate for the Neo 4000. Drawing of the next frame starts while the previous one is still being composed, so frames overlap.

Swarm

Many small jobs instead of a few big threads

HOW

A program doesn't create threads by hand. It tells Neo Kit “this can be done in parallel”, and Swarm cuts it into small jobs and distributes them to free cores. A core that finishes takes a job from its neighbour.

THE SAME QUEUE FOR THE CHIPS

A job can carry a tag like “best on Newton” or “best on Neuron”. Swarm sends it there if the chip is free and the job is big enough, otherwise the processor does it.

FOR DEVELOPERS

The same code runs on the Neo 500 with few cores and on the Neo 4000 with many, just faster. There is no manual data transfer, because everything shares one memory. Ported Linux programs (WebKit, LibreOffice, games via Bridge) keep their threads, and the Dispatcher sorts them into classes.

SIMD

MMX's successors for small jobs

WHAT

NEON and SVE2 do the same operation on many numbers at once, and SME multiplies matrices. That is what MMX and SSE did on the PC.

WHEN THE PROCESSOR, WHEN A CHIP

Sending a job to a chip costs a few microseconds. A short job (a sound effect, a small picture, encryption, game maths) finishes on SIMD before it would even reach the chip. A long stream (a film, a scene, physics of a whole world) goes to the chip.

THE SYSTEM DECIDES

Swarm measures and learns where each job is faster, so the developer doesn't have to choose.

Cores

Big and efficient, without hyper-threading

PROPOSAL

4 to 8 big ARM cores and 4 efficient ones. Responsiveness depends on the strongest core, and throughput on the number of cores.

HYPER-THREADING

Hardware hyper-threading, where one core pretends to be two, isn't needed: it gives little and makes work less predictable. Software threads and jobs are the basis of everything.

SATELLITE

MMX, SSE and AVX remain important only on Satellite x86 and in translating old programs and games.

Lesson

What the Cell in the PlayStation 3 taught us

IDEA

One main core and eight helper SPE units (seven active in the console), each with only 256 KB of its own memory. Enormous power on paper.

PROBLEM

The programmer had to move data into those small memories by hand, so development was hard and slow. The next console switched to an ordinary processor.

NEO

Multitasking

Four classes and one rule

1. UNDER YOUR HAND

Typing, the mouse and the window you are touching have a reserved core and part of the GPU. From touch to picture, at most one frame.

2. HEARD AND SEEN

Music, film and calls have guaranteed time on Orpheus and Panorama, so they never stutter.

3. BACKGROUND

Rendering, compiling and backups get the rest and stop the instant you need something.

4. ASLEEP

Full throttle

The maximum for pro programs and games

TURNING IT ON

The Turbo button, or a game or pro program requests it itself.

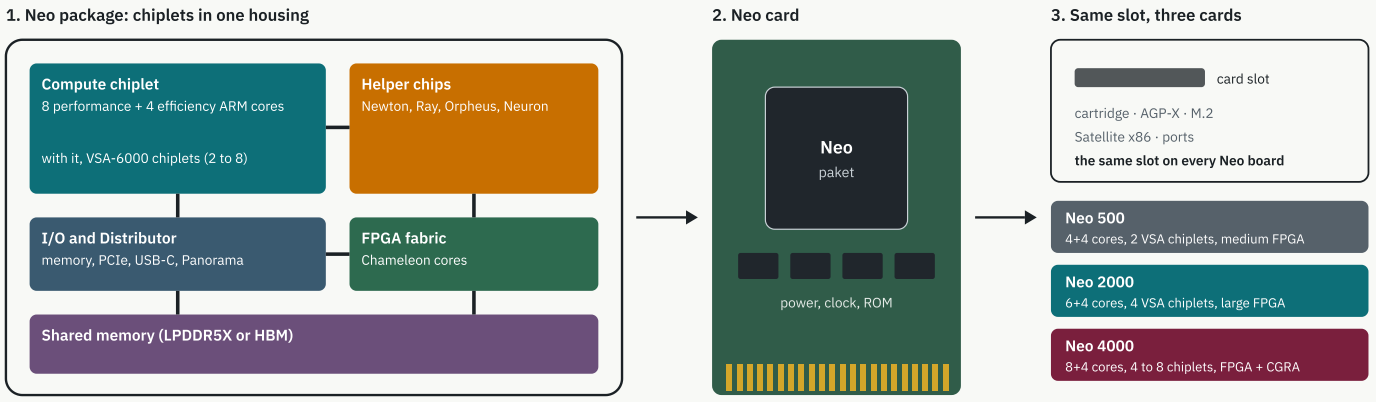
WHAT IT GETS

The big cores, the GPU and part of the cache all to itself (the ARM partitions this in hardware). Everything else moves to the efficient cores, and notifications stay quiet.

NIGHT SHIFT

A long render runs when the computer is idle, or is

Everything on one chip: from the Neo package to a swappable card



The thick lines are UCIe links between chiplets. Small chiplets = better factory yield.

One connector: PCIe, memory for the FPGA, displays, sound, USB, power.

Upgrade = swap the card. The old card can go to the Co-op.

Concept and estimate. Phase 1: off-the-shelf ARM SoC and FPGA on a card. Phase 2: our own Neo package at 12–5 nm.

SoC

Everything important in one package

WHAT

Processor, graphics, Newton, Ray, Orpheus, Neuron, FPGA and memory are in one package and share one memory.

GAIN

Lower latency and lower power, and Swarm doesn't have to copy data. The board has far fewer parts, so it is cheaper and has fewer manufacturing defects.

HYPER-THREADING

We don't add them here either. Silicon area is better spent on extra efficient cores.

Chiplets

Several small chips instead of one huge one

WHY

One big chip has a worse factory yield and is expensive to change. Chiplets are connected with the open UCIe standard and each is made on the process that suits it.

COMBINATIONS

The same chiplets make all three models: 4+4 cores and two VSA-6000 chiplets for the Neo 500, up to 8+4 cores and eight chiplets for the Neo 4000.

FPGA

The FPGA is a separate chiplet, because it is hard to make on the same silicon as the processor.

Card

Drop-in upgrade

HOW

The Neo package, memory and power delivery are on the Neo card with one connector. The motherboard carries the ports, cartridge, AGP-X, M.2 and Satellite, and every Neo board has the same card slot.

PHASES

Phase 1 is an off-the-shelf ARM SoC and FPGA on a card, without our own chip. Phase 2 is our own Neo package on the same board.

FOR THE BUYER

An upgrade is a card swap, so a Neo 2000 can get a 4000-class card. The old card can keep working in the Co-op or a club.

Realistically in a day: one main job, two or three visible windows and a few quiet background jobs. A document, music and a video call together use only a few percent of the processor, because sound and picture go through Orpheus and Panorama. This is a concept estimate, not measured.

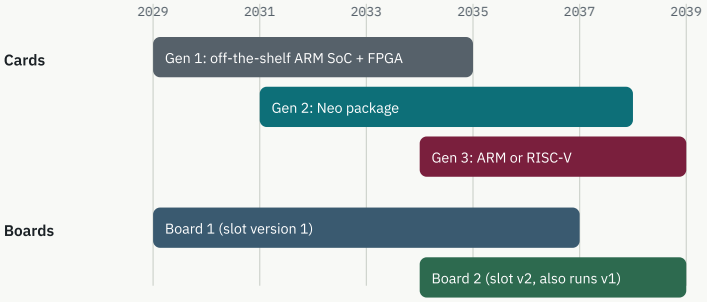
OPEN PLATFORM

Upgrades and expansion: a computer that grows with you

Neo isn't thrown away after three years. The card with the Neo package is swapped through generations, while the board, case, disks and peripherals stay. Expansion is open as on the Amiga: a card says who it is and what it can do, and the schematic and slot are public, so anyone can make add-ons.

Generations of cards and boards, and everything that can be added

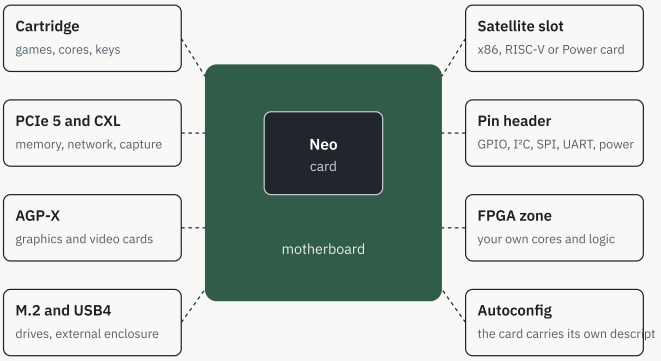
Generations: the card changes, the board stays



Promise: the same slot for at least 8 years.
A new card works in an old board, and a new board takes an old card.
Neo OS from ROM recognises the card and adapts by itself.

Years are approximate. Which card Neo OS 1.0 ships on is still being decided.

Expansion: open like the Amiga, broader than the PC



The schematic, slot and Autoconfig are open (CERN-OHL-S).
Anyone can make cards without permission.

Generations

A slot that lasts eight years

RULE

Every new card works in an older board, and a new board takes an older card. When the slot changes, the new version still takes the old one.

HOW

Neo OS is in the card's ROM, so with a new card you also get a system that knows it. Settings and RAD: stay on the board and disk.

OLD

The old card is taken back at a discount, goes to the Co-op, a club or a school, or becomes a small server.

Autoconfig

The card says who it is

IDEA

Like Zorro on the Amiga: every card carries its own description, icon, signed driver and help. Plug it in and it works, without hunting for drivers.

SAFE

The driver runs in its own Fortress cell, and the IOMMU keeps it away from memory that isn't its own.

OPEN

The description format, slot and schematic are public (CERN-OHL-S). No permission is needed to make a card, and the Neo test suite grants the “works with Neo” mark.

Beyond the PC

What the PC doesn't have

SATELLITE SLOT

It isn't only for x86. The same card can carry a RISC-V or Power processor, for learning, old programs or work on another architecture.

PIN HEADER

Protected GPIO, I²C, SPI and UART on the front of the case, with a fuse. Sensors, robots and school projects from Lux and the Workshop, without a separate board.

FPGA ZONE

Part of the FPGA fabric is yours: your own logic, your own core or accelerator, with templates in the Workshop.

CARTRIDGE

Games, cores, keys and tools, as on old consoles, but at NVMe speed.

Which processor?

ARM now, RISC-V as the next step, PPC on a card

OUR OWN CORE?

No. With an architecture licence it costs about €0.5 to 1 billion and 4 to 5 years, and pays off only at millions of units. The feel in use comes from our chips and the OS.

ARM

The main processor for Gen 1 and 2: the strongest single core per watt, mature software and good translation of x86 games.

RISC-V

No licence, and Fortress (seL4) already runs on it. Since the compute chiplet is replaceable, a Gen 3 card can be RISC-V on the same board.

POWERPC

There is no modern efficient desktop PPC chip. So PPC goes on a Satellite card for AmigaOS 4 and MorphOS (the buyer obtains the licence) and into emulation on ARM. The classic Amiga goes through the FPGA core.

Options

Cheap things that mean a lot

VIDEO INPUT

HDMI input: old consoles and other computers go through Panorama's processing and recording. About €25.

ON THE CASE

A volume dial and button with a mute LED (€5 to 40). An e-ink strip for the calendar, the Board and music (about €20).

PRESENCE SENSOR

Without a camera (ToF): locks and dims the screen when you leave. About €3.

FOR PRO AND GAMES

Colour calibration in the Neo monitor (about €15), a light gun for modern screens (about €30) and a quiet cooling mode.

Repair

Open, replace, carry on

CASE

Ordinary screws, no glue. The card, fan, power supply and battery are replaced without tools or with one screwdriver.

PARTS

The schematic, parts list and manual are public, and parts are sold for at least 10 years.

UNDER THE HOOD

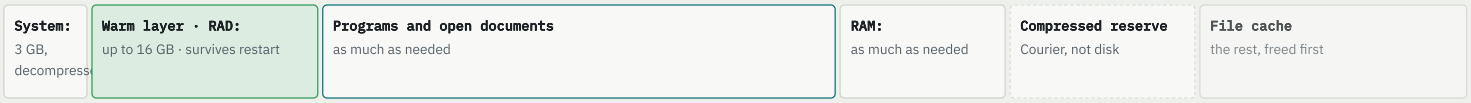
Shows which card it is, how long it has been running, the temperature and when it's time to clean it.

All years and prices are concept estimates. The promise about the slot and parts is a proposal for the Neo consortium.

LIVING MEMORY

No swap, no waiting, no loss

Neo never moves memory to disk. RAM:, RAD: and programs share one compressed space that divides itself as needed. Everything important for speed and every open document survive a warm restart, so after a restart everything comes back in a second. Sizes are a proposal for a Neo 2000 with 64 GB.



The limits aren't fixed: while RAM: and RAD: aren't using space, it serves programs and the cache. The warm layer is marked in green because the memory controller preserves it during a warm restart.

When memory runs out, in order	What happens	The user notices
1. File cache	Freed first, because it can be read again.	Nothing
2. The fast layer of RAD:	The part of RAD: with pre-prepared programs and libraries shrinks. Live copies of open documents are never touched.	Nothing
3. Program code	Programs from ROM and neopak packages are read-only, so their code is released and reloaded when needed. No swap is needed for them.	Nothing
4. Compression	Rarely used memory is compressed in RAM via Courier, in hardware, about 2 to 3 times smaller.	Almost nothing
5. Putting to sleep	The program you haven't used the longest is frozen, and Chronos writes a compressed snapshot of its whole state into the warm layer. A click wakes it in a fraction of a second, in the same state.	The program icon gets a “sleeping” badge
6. Asking	If even that isn't enough, Neo says which programs take up the most and offers to close some. It never slows down the whole system.	One message, instead of freezing

Warm layer

What survives a warm restart

DOCUMENTS

Every open document, with unsaved changes, undo history and cursor position. Every change is immediately written to a journal in RAD:.

PROGRAMI

The state of open programs and windows, the clipping Collection and the loaded FPGA core.

SPEED

The decompressed system core, the shader and font cache, the search index, shared libraries and the list of programs you use most, already prepared.

Durable

RAM speed, disk safety

WRITING

Every change goes to RAD: immediately, and Courier copies it compressed to the SSD within a couple of seconds. Saving is therefore instant.

POWER LOSS

The Neo 500 has a battery. The Neo 2000 and 4000 get a supercapacitor in the power supply, which gives a second or two to flush the last writes to the SSD.

COLD BOOT

Courier restores the warm layer from the SSD with hardware decompression, in two to three seconds for a few gigabytes.

Dynamic

RAM: and RAD: grow and shrink

RAM:

Temporary, erased at restart. Takes up only as much as is in it.

RAD:

Survives a restart, has a permanent copy on the SSD and a limit per model: 4, 16 or 32 GB.

FAST LAYER

The Dispatcher tracks what you use and keeps the most used programs and libraries in the fast layer of RAD:. Films and large archives stay on the SSD.

Hardware

What the machine must have

CONTROLLER

A memory zone in self-refresh during a warm restart, which Neo ROM doesn't erase.

INTEGRITY

ECC and periodic memory scrubbing, because data lives in RAM for a long time. A per-block checksum in the warm layer.

PROTECTION

The warm layer is encrypted with a key from the Safe chip. Network keys and passwords aren't kept and are asked for again.

Estimate, not measured: first program launch from about 300 ms to under 100 ms, return after a warm restart from about 10 s to about 2 s, with all windows and documents in place. In already warmed-up work the gain is 10 to 20%, but there are none of the stalls swap causes when memory fills up, and saving is instant.

EYE

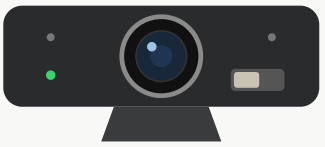
A camera included with every model

Every Neo comes with a camera, and the class of camera grows with the model. Quality is built on three things: a larger sensor, glass optics from a partner from the photography world (a Zeiss- or Leica-type manufacturer, under a licence agreement) and image processing in Neuron. All three cameras have a mechanical shutter and are connected to Lock.



Eye S

with the Neo 500



Eye M

with the Neo 2000



Eye Pro

with the Neo 4000

Light dot: LED tied to the sensor power · slider: mechanical shutter

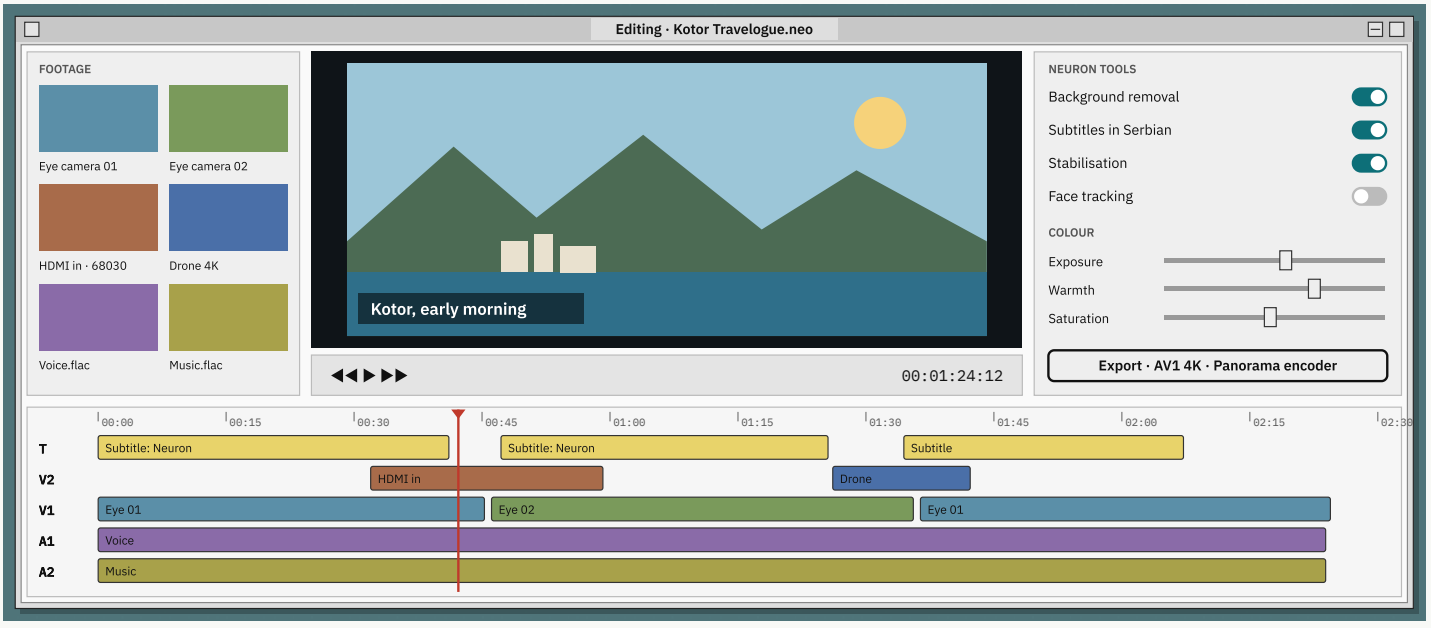
	Eye S	Eye M	Eye Pro
Sensor	1/2"	1/1,3"	1", large pixels
Optics	Glass, fixed focus	Glass, autofocus, partner optics	Partner optics, bright, with a thread for swapping lenses
Video	4K 30	4K 60, HDR	4K 120, 10-bit log recording
Depth	Neuron only	Neuron only	ToF sensor + Neuron
Microphones	2	4, directional	4 + input for an external microphone
Connection	USB-C	USB-C	USB-C, raw stream directly into Panorama
Privacy	Mechanical shutter, an LED that software can't turn off, a physical switch on the computer case		

EXAMPLES OF USE

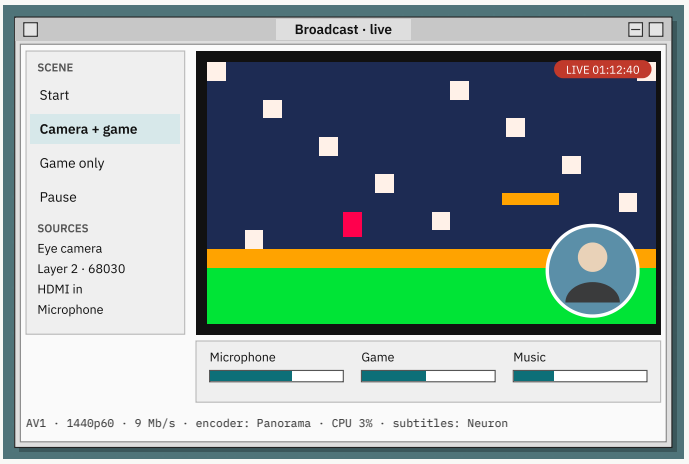
Apps at work

Screens in the Neo Platinum style. The content is an example.

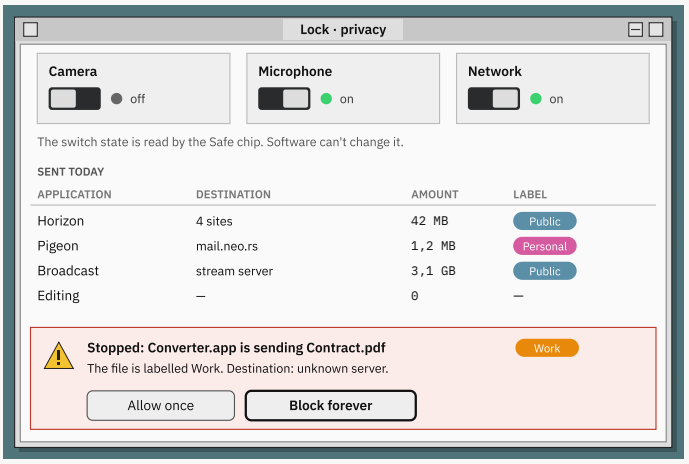
Montage: video editing with Neuron's help



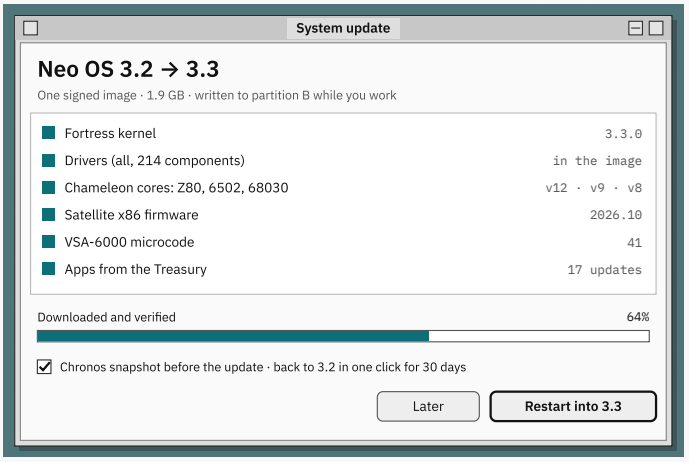
Broadcast: streaming a game from a retro core



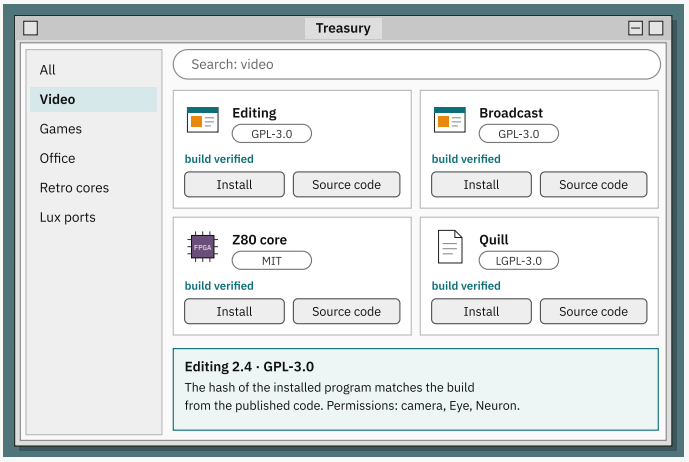
Lock: who sends what from the machine



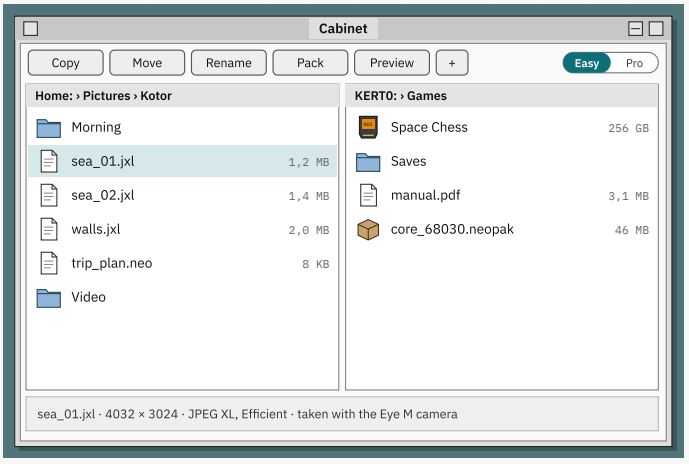
Renewal: one update for the whole system



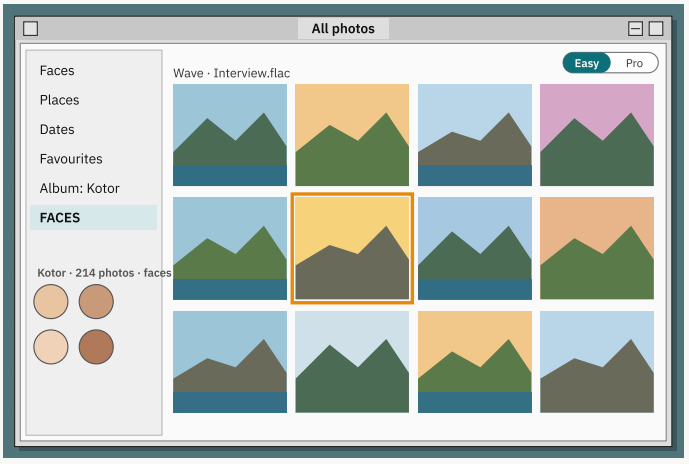
Treasury: open apps with verification



Cabinet: files in two panels



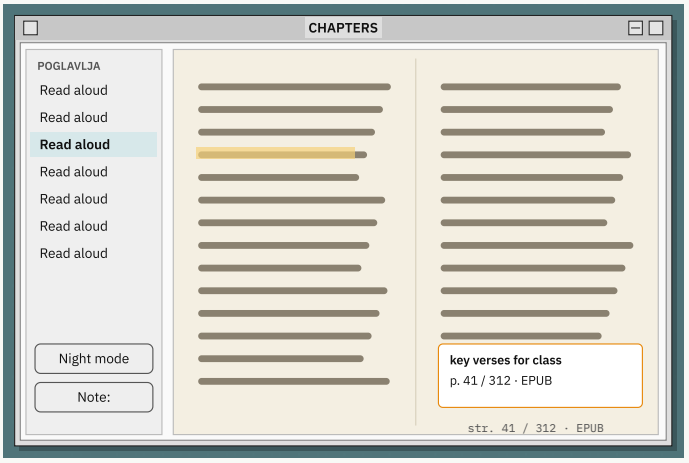
Album: thousands of pictures at your fingertips



Wave: sound editing



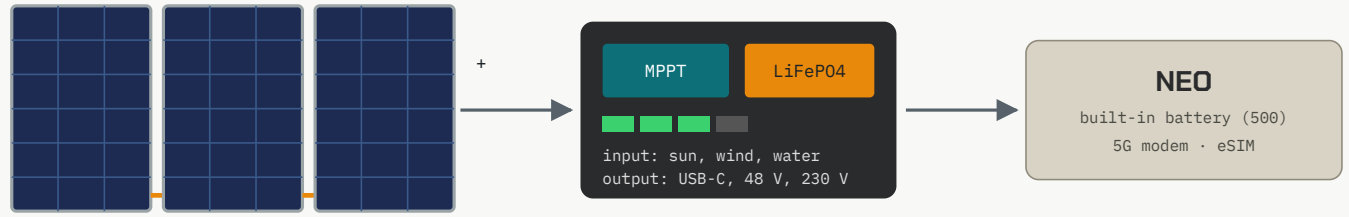
Leaf: books and PDFs



SUN

Independent power

External 100 W solar panels connect in a chain, so the system grows along with the model's consumption. The charge controller and battery are in the Sun station, and with the 5G modem the computer works without mains power and without WiFi. Figures are concept estimates, without a monitor (add 15 to 30 W).



Sun panels · 100 W each · connect in a chain
foldable · USB-C PD or MC4

Sun station
300 Wh · 1 kWh · 2 kWh, stackable

Energy in Neo OS
watt display, power-saving mode, FPGA shutdow

In winter Serbia gets about four times less sun than in summer: a larger array or power-saving mode is needed then

	Neo 500	Neo 2000	Neo 4000
Consumption	10 to 35 W	40 to 150 W	80 to 400 W
6 hours of use per day	About 150 Wh	About 500 Wh	About 1.2 kWh, about 600 Wh in power-saving mode
Panels in summer	1 × 100 W	3 × 100 W	6 × 100 W
Battery	Built-in 70 Wh + 300 Wh station	1 kWh station	2 kWh station
Without sun or mains	All day	About one day	One evening to one day

Other sources and energy saving

Generation	What it gives	Rating
Small 300–400 W wind turbine	Charges the station at night and in winter too, when there is no sun. A good complement to panels at a weekend cottage.	Worth it
Micro hydro on a stream	If there is a stream, it gives a constant 100 W or more, day and night. The best source where it is possible.	For special cases
Pedal or hand dynamo	A pedal gives 50 to 100 W while turning, and a hand crank 10 to 20 W. Enough for a Neo 500 in an emergency.	For special cases
Thermoelectric module on a stove	Turns the heat of a wood stove into 10 to 30 W. In winter it covers exactly the gap the panels leave.	For special cases
Piezo in the keyboard, radio waves	They give milliwatts, thousands of times too little for a computer.	Not worth it
Saving	**What it gives**	**Rating**
Sunny mode in Neo OS	Heavy jobs (rendering in Montage, Renewal, backups) wait for surplus sun, like a smart grid in miniature.	Worth it
Turning off chiplets	The Dispatcher fully turns off VSA-6000, the FPGA and Satellite when they aren't needed. On the desktop only the ARM runs.	Worth it
The right job on the right chip	Video encoding goes to Panorama, AI to Neuron, retro to the FPGA. A dedicated block uses ten times less than the CPU.	Worth it
Chronos instead of sleep mode	The computer turns off completely (0 W) and comes back in a second or two, so shutting down costs nothing.	Worth it
48 V DC input	Power comes straight from the station, without converting to 230 V and back. That's 10 to 15% less loss.	Worth it
Passive cooling	The Neo 500 has no fan. In the larger models the fans run only under load.	Worth it
E-paper monitor	For writing and reading in the sun, uses almost nothing. A nice option for the portable Neo 500.	For special cases
Heat as heating	All the power of a Neo 4000 ends up as heat. In winter the case can blow into the room, in summer outside.	For special cases

All additional sources go into the same MPPT input of the Sun station, so they combine without extra equipment.

NEO GAMES

Games included and on cartridge

Every game shows off part of the hardware. Where a well-known original exists, the game is either licensed or an original game of the same genre, with its own characters, rules and look. The covers are sketches.



Grandmaster

Chess · Original, Stockfish engine

A strong opponent with a human playing style and lessons for beginners. 3D board and game analysis.

Shows off: Neuron, VSA-6000

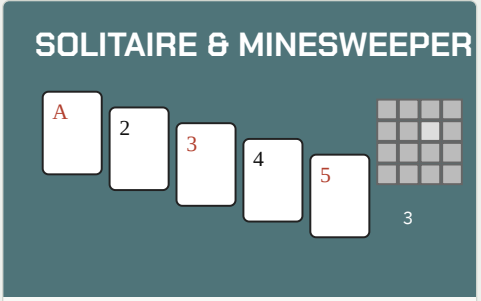


Midnight Poker

Card game · Original · 18+

Cheeky poker in an illustrated style, for adults. It unlocks only on a profile with verified age, and a parental profile doesn't see it.

Shows off: profiles and the Lock



Solitaire and Minesweeper

Classics pack · Original, Windows-classics genre

Solitaire, FreeCell, Hearts and Minesweeper in the Neo Platinum look. On the computer from the first power-on.

Shows off: Neo Kit and the GUI

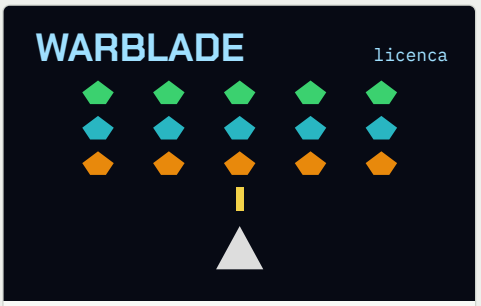


Cosmos Pinball

Pinball · Original

Space pinball with real ball physics and several tables.

Shows off: Newton, physics

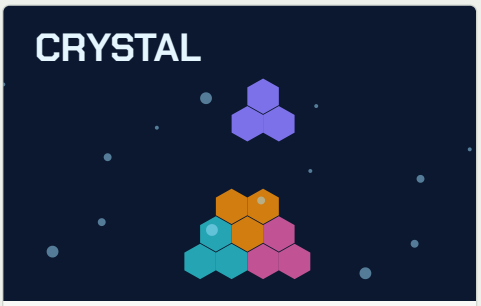


Warblade Neo

Shooter · Licence, rights holder

A licensed game from the Amiga era. The original runs in the 68030 core, and the new version gets HD graphics.

Shows off: Chameleon, cartridge



Crystal

Puzzle, falling blocks · Original

Hexagonal blocks, with sound and particles tied to the music. The rules are our own, because Tetris is protected by licence.

Shows off: VSA-6000, Orpheus



Forest Bane

Platformer · Original

A boy whom a curse turns into animals, and each one opens new parts of the world. Colourful 2D graphics.

Shows off: Copernicus, raster effects

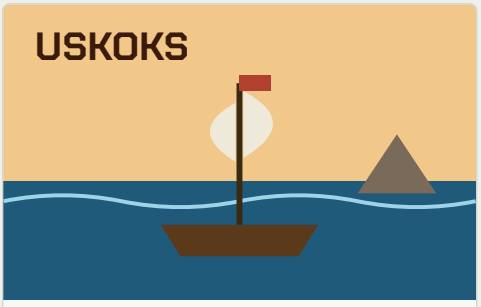


Block 70

First-person shooter · Original, Quake engine (GPL)

A fast retro shooter among the concrete blocks of an alternate New Belgrade, 1986. Pixelated textures, fast movement, secret rooms.

Shows off: VSA-6000 and Glide mode

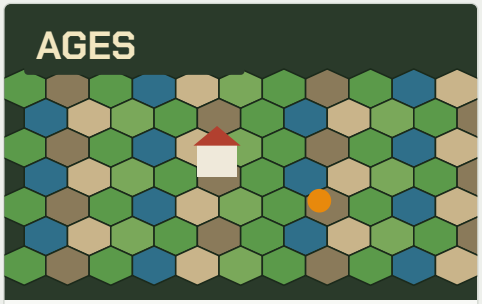


Uskoks

Naval adventure · Original

The Adriatic in the 16th century: trade, sailing, ship battles and swordplay. An open world and your own career from sailor to captain.

Shows off: Newton (sea and sails), Ray



Ages

Turn-based strategy · Original, Freeciv engine (GPL)

Lead a people from the Stone Age to space. Diplomacy, science and building; a match for one evening or for a whole month.

Shows off: Neuron (opponents), Together



Quiz Night

Party game · Original

A quiz for 2 to 8 players in the same room or over the network, with the Eye camera for the players' faces. Users write the questions themselves.

Shows off: Together, Eye



Metroidvania

Action, exploration · Port licence, e.g. Hollow Knight

Talks with the authors about an ARM port. The fallback is our own game “Ras”, set in medieval Serbia.

Shows off: Chronos, continue anywhere

Licensing proposal: “Ide zmija” by DJ Nick (Deetronic Design Studio), today a running game for the browser and Android, as a Neo 2D platformer with the Snake as the hero,

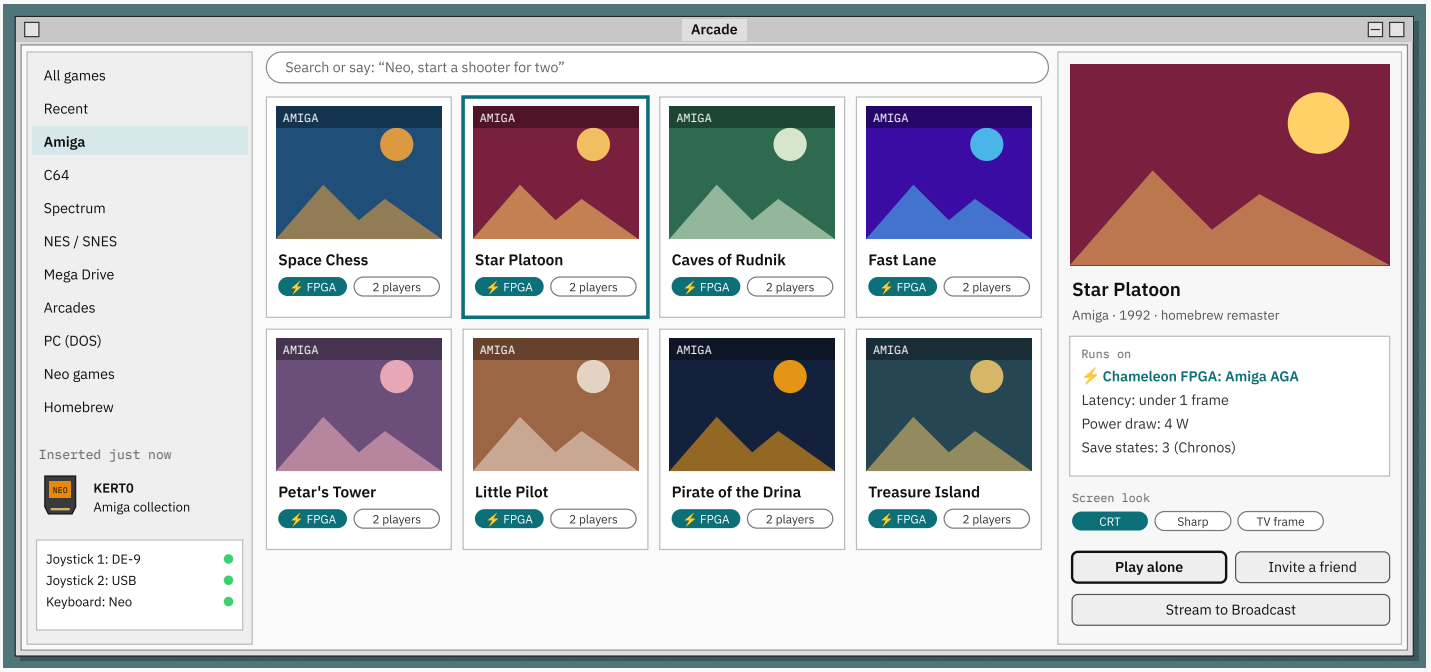
the song's music and Newton physics. It goes with the Crystal style, only under a contract with the author.

ARCADE

All the old games in one place

Insert a cartridge, disk or file, or pick a game from the shelf. Neo itself chooses how to run it, and the user doesn't need to know anything about cores and emulators.

Arcade: a game shelf for every system



How Neo chooses the path in the background



Same for everything: controls, save states, rewind, CRT look, network play

FPGA core

Like the original hardware

FOR WHAT

Amiga, C64, Spectrum, Atari, NES, SNES, Mega Drive and old arcades.

WHY

Exact timing, latency under one frame and low power, which is good for the Neo 500 on battery and solar.

Emulator

Thousands of arcades and 3D consoles

FOR WHAT

MAME and libretro cores, for systems where an FPGA isn't practical.

ACCELERATION

VSA-6000 renders 3D at a higher resolution, and Neuron helps with image scaling.

Box

Old PC programs

FOR WHAT

DOS and Windows games through Satellite x86, Wine/Proton or a ReactOS box.

LOOK

The window behaves like any other Neo window, through Clasp.

Same for all

One set of controls, one menu

FEATURES

Save states (Chronos), rewind, CRT look, Turbo button, network play and streaming to Broadcast.

CONTROLS

Every joystick, DE-9 or USB, works in every game without setup.

Legal

Your games, fairly

ROMS

Neo doesn't ship other people's commercial ROMs. A USB reader turns your original cartridge into a file.

FREE

MAME ROMs the authors have allowed, homebrew games, licensed collections in the Treasury and the AROS ROM instead of Kickstart.

Amiga core

An Amiga in every Neo

BASIS

The open Minimig-AGA cores (GPL) and the TG68 processor (LGPL).

TURBO

A fast 68080 with the SAGA chipset as a paid add-on, if a deal with the Apollo team is reached.

TOGETHER

Playing and working with other Neo users

Network play is a recommendation for every game in the Treasury. Collaboration is built into Quartet, Montage and Neo Cards. The connection goes directly between computers, encrypted, without a mandatory cloud server.

Play

Every game multiplayer

RECOMMENDATION

The Treasury marks games that have network and same-room play. Neo Kit provides a ready-made system for finding players and rooms.

CHRONOS

Chronos state snapshots enable “rollback” netcode: the game feels local even when the opponent is far away.

RETRO

Two Neo computers play the same game from a 68030 or Z80 core together, as if the joysticks were in the same machine.

Work

Together on the same file

DOCUMENTS

Quill, Abacus and Stage: several people write in the same document at the same time, and changes merge by themselves (CRDT).

VIDEO AND CARDS

A shared timeline in Montage and shared Neo Cards decks.

SCREEN

Screen sharing and Help a Friend: someone shows you how to do something, with your permission, which is withdrawn with one click.

No intermediaries

Yours, not ours

CONNECTION

Directly between computers, even on a local network without the internet. The server only helps computers find each other.

LOCK

The Personal and Work labels apply here too: Lock asks before a file is shared.

CLUBS

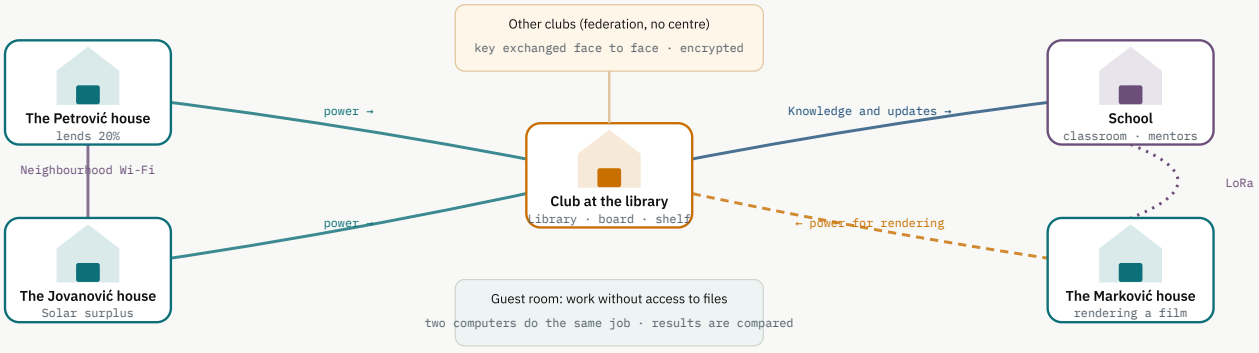
User communities can set up their own discovery server, like the BBSes of old.

CO-OP

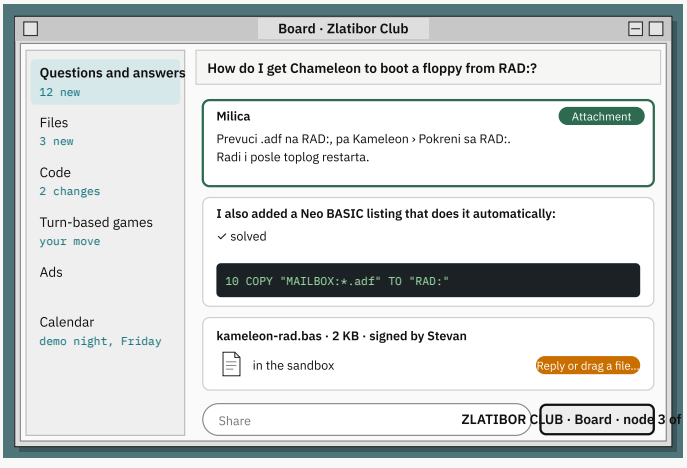
Shared knowledge, borrowed power, clubs without a centre

The spirit of the old computer clubs and free software, built into the system: users share knowledge, results and part of their computer's power, voluntarily and without intermediaries. It all rests on the same foundation as the rest of Neo, namely Keys (permissions) and closed rooms, so sharing never opens your files.

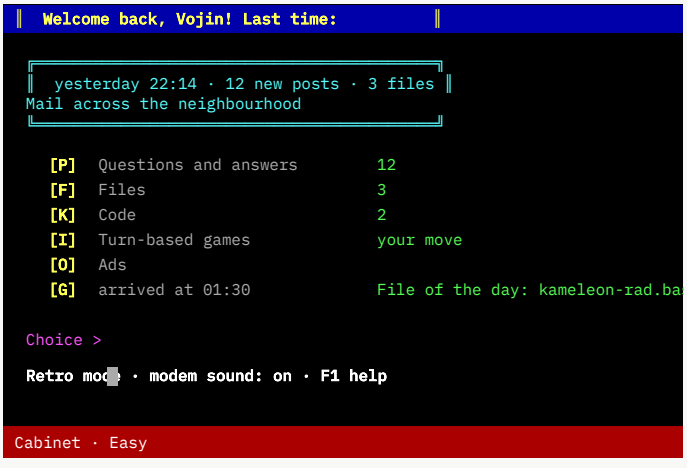
A neighbourhood club: homes, a school and a library share knowledge and power, even without the internet



Board: ads, files, solutions and code in one place



The same Board in retro mode, like a BBS from the nineties



Lending power

A computer that helps while it's idle

WHAT

An idle Neo lends part of its power to a friend, a club or science: rendering a film, translating a book, building a program, training a local model or a science project of your choice.

WHEN

Only when it is plugged in, when you aren't using it and, if you want, only from the Sun station's surplus. You set the limit for power, hours and temperature.

REWARD

Not money but thanks: whoever helped gets priority when they need the club's power.

Guest room

Someone else's job, your data untouched

SANDBOX

The borrowed job runs in a closed room on Fortress, without access to files, the camera, microphone or the rest of the network. It gets only the input, it returns only the result.

CHECKING

The same piece of work is done by two computers, and the results are compared. A wrong or fake result is seen at once.

Shared results

Done once, available to all

WHAT IS SHARED

Shader caches for games, OCR and translations of public domain books, map tiles, Neo packages and updates to the Knowledge cartridge, directly between users.

TRUST

Programs are built reproducibly, so anyone can check that the same source code gives the same file. A result is valid when the fingerprint matches from several independent sources.

Code

See how it works

SOURCE

Every Neo app has a “See how it works” button that opens its source code in Neo Kit. A change can be tried, saved as your own branch and shared with the club.

LISTINGS

Neo BASIC programs are shared like listings from old magazines: one file, runs right away, and you see right away how it was made.

Klub

Cyber computer club

WHERE SHE LIVES

On one Neo at someone's home, in a library, school or cultural centre. Clubs connect with each other without a central server, like a federation.

WHAT IT HAS

A shelf of programs, games and levels, its own Library, shared power, a message board in the spirit of a BBS, a calendar of meetups and demo competitions in Arcade.

PEOPLE

Mentors help newcomers through Help a Friend. The club chooses its own rules and moderators.

Neighbourhood

Even without the internet

NETWORK

Neo computers in the neighbourhood connect directly over Wi-Fi, and at a distance over LoRa radio for messages and small files, for villages and schools without good internet.

CARTRIDGE

Big things, like the whole Library, travel on a cartridge from hand to hand, signed so it's known they weren't changed.

Board

A BBS built into the system: sharing like network play in every game

EVERYWHERE

Just as every game in the Treasury has network play, every app has “Share with the club”: a document, a picture, a game level, a Neo BASIC listing, settings or an Arcade core. The file goes to the club Board in one move, with a description and licence.

SECTIONS

Questions and solutions (the answer that helped gets a “solved” badge, so Dijana sniffs it out the next time someone asks the same thing), Files, Code with change review and branches, turn-based Games like the BBS doors of old, and Ads for trading equipment.

MAIL ACROSS THE NEIGHBOURHOOD

Like the old BBS networks: messages and files travel from club to club when the computers meet, over the internet, the Neighbourhood network, LoRa radio or a cartridge. It works even where there is no internet.

LOOK

A modern view in Neo Silver, or a retro mode with text and ANSI colours, with modem sound if you like. Same content, you choose how it looks.

SAFETY

Every file is signed with the author's key and verified before opening. A program from the Board runs in a sandbox until two members sign it. The Board shows no ads and doesn't track who reads what.

Security

Share, and stay protected

MEMBERSHIP

You join by invitation, and keys are exchanged face to face via a QR code, like a handshake. Trust is built through people you already know.

SOMEONE ELSE'S CODE

A program from the club runs in a sandbox until two members review and sign it. Only then can it ask for wider permissions, and even then only with your consent.

CONNECTION

Everything is end-to-end encrypted. Lock's Personal and Work labels apply here too, so the system asks before anything labelled leaves the computer.

KIDS

A child profile joins only clubs the parent approves, without power lending and without messages from strangers.

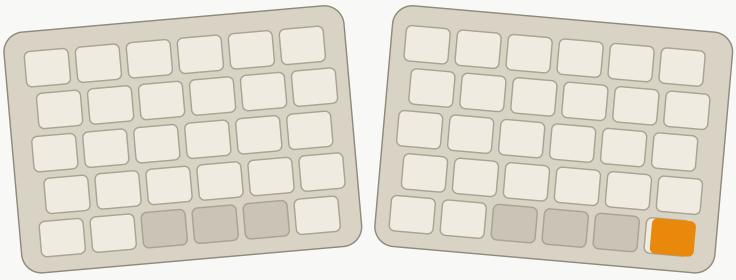
FREEDOM

Whoever leaves the club takes their data and programs with them. No account with us, no tracking and no ads.

PERIPHERALS

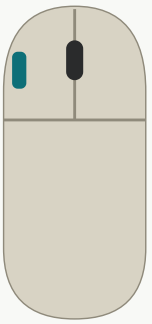
Mouse, touchpad, keyboard and remote

A quality mouse and a small touchpad come with every model. The ergonomic mechanical keyboard and the remote cost extra. Every Neo also works with ordinary USB and Bluetooth peripherals.



Ergonomic mechanical keyboard (extra cost)

split, tilted, dual Cyrillic/Latin legends, Voice key



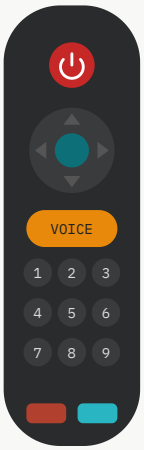
Neo mouse

3 buttons, scroll wheel

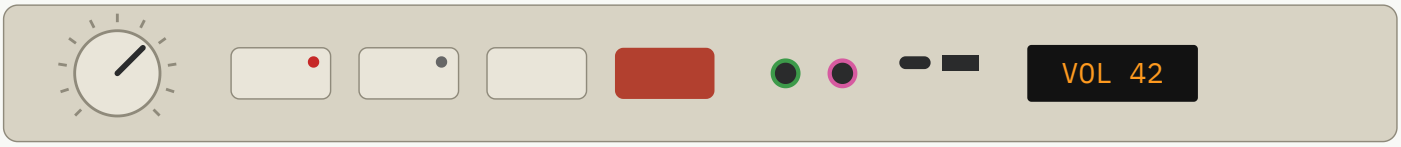


Touchpad

multi-finger gestures



Neo remote



Volume · Microphone · Quiet · Voice · Chronos · Headphones · microphone · USB · Display

In the box and extra

Peripherals

IN THE BOX

A Neo mouse with three buttons (the middle one opens the menu, like back then) and a small touchpad for multi-finger gestures.

EXTRA COST

An ergonomic split mechanical keyboard with quiet switches, dual legends and a Voice key.

ANY

Ordinary USB and Bluetooth peripherals work right away, without drivers.

Case

Buttons that always work

SOUND

A round volume dial, like on an old radio, and a mute button.

MICROPHONE

The button cuts the microphone in hardware, and a red light confirms it.

OTHERS

Voice, Chronos keys and front headphone and microphone jacks. The display shows the volume while you turn the dial.

Remote

Neo remote

CONTROL

Arrows, numbers for teletext pages and channels, and when you move it in the air it works as a mouse.

VOICE

Hold the Voice button and say a command. The microphone works only while the button is pressed.

GAME

Turned sideways it becomes a joystick for retro games. It also has Chronos and Retro buttons. A phone can replace it via Link.

Voice

Speech to text and text to speech

SPEECH TO TEXT

Dictation into any field, in Serbian, in Latin or Cyrillic, locally on Neuron.

COMMANDS

“Open Album, show Kotor.” Voice is translated into Lux commands and works with all apps.

TEXT TO SPEECH

A natural Serbian voice reads books in Leaf, messages, notifications and the whole screen for the visually impaired. A retro robotic voice is an option.

Talking with Neo

The whole computer by voice, and it answers

HOW

Speech to text, Ask Neo and text to speech work together: you say what you want, Neo does it and answers by voice. All locally on Neuron. Claude for harder questions is an option you turn on yourself.

EXAMPLE

“Neo, how much free space do I have?” → “212 gigabytes. Most of it is taken by the videos from Kotor.” → “Move them to the external disk.” → “Moving 38 files. Done in about two minutes.”

SAFETY

Before an action that can't be undone Neo asks out loud and waits for a “yes”. Deleting goes to the Trash, and everything else is undone with the word “undo”.

NO SCREEN

A voice-only mode for the blind and visually impaired, for the kitchen and for driving: Neo reads the menu, windows and notifications and guides you through every app.

PRIVACY

Neo listens only while you hold the Voice button, or after the wake word if you turn it on. A light shows when it is listening.

PEOPLE'S COMPUTER

A computer made to human measure

A computer that can be repaired, that lasts, speaks both scripts, works without the internet and asks for no subscription.

Repair

Repaired at home

PARTS

Ordinary screws, a replaceable battery, keyboard and fans. Parts available for ten years.

MANUAL

A repair manual with pictures is in the OS itself, and the schematics are public.

Longevity

A computer for ten years

RENEWAL

At least ten years of updates. Every old model runs the new OS, and Efficient mode helps the weaker ones.

UPGRADE

A new board fits into the old case, so only what is outdated gets replaced.

Price

For every pocket

NEO PEOPLE'S

A Neo 500 without extras, targeted under €400. A refurbished official Neo with a warranty for even less.

PAYMENT

Trade-in of the old computer, interest-free instalments and Neo in libraries and schools for borrowing.

Language

Cyrillic and Latin as equals

TRANSLITERATION

One switch shows the whole system and every document in the other script, instantly and without errors.

KEYBOARD

Dual letter legends on the keys, and spelling and speech work in both scripts.

Knowledge

Works without the internet too

LIBRARY

Wikipedia, dictionaries and manuals preloaded and available offline (Kiwix).

SCHOOL

Neo BASIC lessons, Neo Cards for projects and the “First Steps” game that teaches the mouse and keyboard.

Family

Profiles for everyone

KIDS

A child profile with a time limit the parent can see, but no tracking.

OLDER PEOPLE

Simple mode: big letters, six big buttons and Help a Friend with one click.

The state

Your ID card in the computer

SIGNATURE

The Safe chip stores a qualified electronic signature, so e-government and signing contracts need no card reader.

STANDARDS

All documents in open formats, which the state can still open in 50 years.

No subscriptions

Bought means bought

SOFTWARE

Everything in the box is paid for once, with the computer. No monthly subscriptions and no ads.

CLOUD

The optional cloud is on a server you choose yourself: at home, at the club or at Neo.

Community

Neo clubs

CLUBS

Neo clubs, repair shops and “repair cafés” across cities, like the computer clubs of old.

MAGAZINE

A Neo magazine with programs to type in, as in the days of the Galaksija.

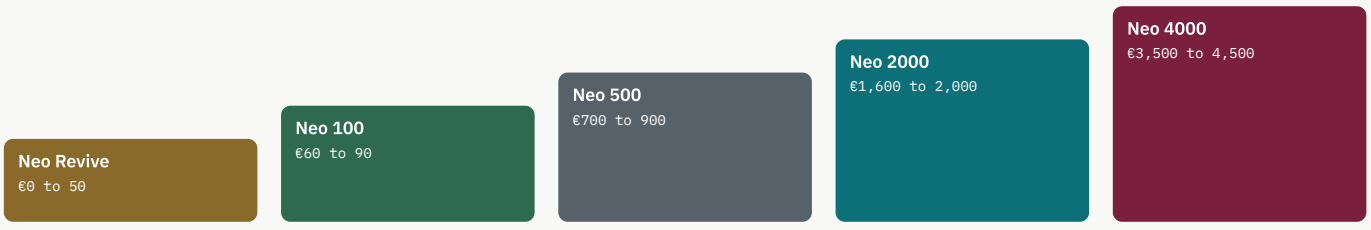
NEO FOR EVERYONE

Neo 100 and Neo Revive: the same spirit for much less money

A Neo 500 and a cheap OLED screen are still too much for much of the world. So the family gets two steps below: Neo 100, a single ARM board in a keyboard, and Neo Revive, the same system on old computers. Both exchange files and work in the same community as the big Neo.

The Neo family from €0 to €4,500

One family, from an old laptop to a tower: the same formats, the same community



Neo Revive	Neo 100	Neo 500	Neo 2000	Neo 4000
old laptop, PC or tablet Neo OS Lite from a USB key	ARM board in the keyboard TV or a used monitor, 5 W	Neo package, FPGA full Neo OS	VSA-6000, Satellite option full Neo OS	everything, up to 8 chiplets full Neo OS

Common to all: the same file formats · Board · Co-op and power borrowing · Mail across the neighbourhood · Offline knowledge · Treasury from a USB key

Neo 100 and Revive borrow power from bigger Neo computers in a club or school through the Co-op, for rendering, translation and big jobs.
Prices are estimates. Neo 100 and Revive run only in the Retro style and without FPGA, and old games go through an emulator.

Neo 100

A computer in a keyboard for €60 to 90

WHAT

One ARM chip of the single-board computer class (4 cores, 4 to 8 GB), eMMC, WiFi, two USB ports, HDMI and composite output. No FPGA, VSA-6000 or Satellite.

SCREEN

Any TV, a used monitor or a cheap 10" IPS panel (about €30). Nothing has to be bought with it.

POWER

About 5 W over USB-C. Runs from a power bank or a small 20 W solar panel.

Neo Revive

An old computer, a new life

WHAT

Neo OS Lite boots from a USB stick on old laptops, PCs and ARM tablets. Fortress (seL4) runs on both x86 and ARM.

WHY

Used laptops cost €0 to 50, and would otherwise become e-waste.

WHERE

Schools, libraries and clubs with an installation stick and a list of tested computers.

DOESN'T RUN EVERYWHERE

Only on tested computers: 64-bit Intel or AMD with UEFI, roughly from 2012, at least 2 GB, Intel or AMD graphics (Nvidia only basic), and a few ARM boards of the single-board computer class. An old 32-bit PC can't. Before installation the stick checks the computer and says whether it will work.

DRIVERS

The only exception to the rule about our own drivers: for old PCs Linux driver code is used, locked in isolated parts of Fortress.

Neo OS Lite

Same kernel, less shine

SAME

Fortress, Neo Kit, file formats, Quartet, Leaf, the Board, Lux and Neo BASIC.

LESS

Only the Retro style and the Calm level, no 3D effects. Old games run through an emulator (8 and 16 bits without problems), not through an FPGA.

EASY

Easy mode is the default, and apps are chosen for 1 to 2 GB of RAM.

Community

Everyone in the same neighbourhood

SHARING

The Board, Neighbourhood Mail and the Co-op work the same as on the big Neo, even without the internet, over WiFi from computer to computer.

BORROWING

The Neo 100 sends heavy jobs (rendering, translation, speech recognition) to a Neo 2000 or 4000 at the club or school, through a Guest room.

KNOWLEDGE

An encyclopedia, maps, books and school on one 64 GB stick or card, copied from stick to stick.

Treasury on a stick

No internet and no card payments

HOW

A school or club keeps a copy of the Treasury on disk, and the others refresh it with a stick. Signatures guarantee that nothing has been changed.

PAYMENT

Everything in the base system is free. For paid items there are vouchers in local currency and via mobile money.

Who pays

Local manufacturing and solidarity

LICENCE

The Neo 100 is open (CERN-OHL-S), so local workshops assemble it themselves and pay only for the chip.

SOLIDARITY

Proposal: 2% of every Neo 2000 and 4000 goes into a fund for the Neo 100 in schools.

LONGEVITY

The same board for at least 7 years, parts and repair at home.

All prices are estimates for a run of 100,000 units a year and have not been checked against quotes. Neo 100 and Neo OS Lite are planned for version 1.1, and Revive for 1.5. Neo 100 and Revive are part of the Neo community on an equal footing with the other models.

KNOWLEDGE IN A BOX

Open knowledge, always at hand

Everything on this page is publicly available and under free licences. Neo gets it on the “Knowledge” cartridge (512 GB), it works without the internet, and Renewal refreshes it once a month. The Library app searches everything at once, and Ask Neo answers from it and cites the source.

Encyclopedia

Wikipedia offline

CONTENTS

Wikipedia in Serbian, Serbo-Croatian and English, with Wiktionary, Wikivoyage and Wikisource, via Kiwix.

LICENCE

CC BY-SA, free to share.

Maps

All of Europe without a signal

CONTENTS

OpenStreetMap maps with navigation on foot, by bike and by car, all offline. Natural Earth for the world atlas.

LICENCE

ODbL and public domain.

Books

A library of classics

CONTENTS

Project Gutenberg, Standard Ebooks and Serbian classics in the public domain: Njegoš, Vuk, Branko, Laza Kostić, Nušić. Leaf displays them, and the Neo voice reads them aloud.

LICENCE

Public domain, CC0 and free licences.

School

A classroom without the internet

CONTENTS

Kolibri with Khan Academy lessons, OpenStax textbooks, PhET physics and chemistry simulations and Neo BASIC lessons.

LICENCE

CC BY and CC BY-NC-SA, free for schools and homes.

Science

Sky and Earth

CONTENTS

The Stellarium planetarium on the Copernicus chip, NASA images, open star catalogues and Wikidata for quick facts.

LICENCE

GPL, public domain and CC0.

Music and sound

Music free to use

CONTENTS

Musopen classical recordings, IMSLP and Mutopia sheet music, Freesound CC0 effects, OpenGameArt for games and Neo sounds. Wave and Montage use them without fear of copyright.

LICENCE

Public domain, CC0 and CC BY, so every file carries a licence label.

Pictures and type

Graphics and fonts

CONTENTS

Wikimedia Commons and Openverse for pictures, and Noto, IBM Plex and other OFL fonts with full Cyrillic.

LICENCE

CC i SIL Open Font License.

Language

Voice, translation and spelling

CONTENTS

Whisper for dictation, Piper for the Neo voice, Argos for offline translation, LanguageTool and Hunspell for spelling, Tesseract for scanning paper into text.

LICENCE

MIT, Apache and LGPL. Common Voice (CC0) is used for a better Serbian voice.

Maker

Fix it and make it yourself

CONTENTS

iFixit repair guides, FreeCAD, KiCad and OpenSCAD with parts libraries, and the schematics of the Neo computer itself.

LICENCE

CC BY-NC-SA, GPL and CERN-OHL.

Programmer

Help without searching

CONTENTS

The complete Stack Overflow, DevDocs, man pages and the Arch Wiki, all offline.

LICENCE

CC BY-SA and GFDL.

Everyday life

Health, food, recipes

CONTENTS

Open Food Facts: the Eye camera reads the barcode and shows the ingredients. Plus Wikibooks recipes and free first aid manuals.

LICENCE

ODbL and CC BY-SA.

Reader

Leaf for books, comics and audiobooks

FORMATS

EPUB, PDF, DjVu, FB2, DRM-free MOBI and CBZ and CBR comics. On a small screen a comic is guided panel by panel, and on a large one shown as a double page.

READING

Night paper, font size and line spacing per book, tap-to-define dictionary, notes and highlights saved in the book itself and exported as text.

ALOUD

Every book can be listened to with the local Neo voice, and an audiobook and its text go together, with the current sentence highlighted.

LICENCE

Engine built on the open source Foliate, Poppler and MuPDF.

Libraries

Public libraries on one shelf

SOURCES

Project Gutenberg, Standard Ebooks, Wikisource, the Internet Archive, Open Library lending, LibriVox audiobooks and old public domain comics from the Digital Comic Museum. Local digitized public domain collections, wherever libraries open them.

STANDARD

Connected

Knowledge built into the whole system

TOUCH

A word in any program leads to Wiktionary, a place to Atlas, a term to Wikipedia, and a number with a unit to the converter. All without the internet, from the Knowledge cartridge.

SEARCH

When Dijana sniffs, she offers a page from the Library

Ask Neo

A smart assistant that doesn't make things up

HOW

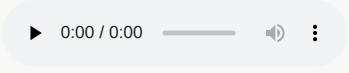
An open language model runs locally on the Neuron chip. It answers only from the Library and gives the source with every answer.

SHARING

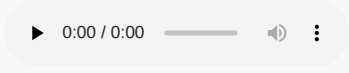
One Neo in a school or village shares the Library over

Neo sounds made by synthesis, CC0

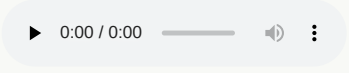
Neo theme



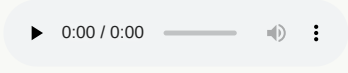
Starting up



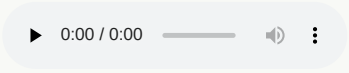
Notification



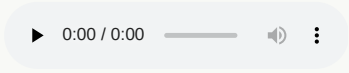
Cartridge



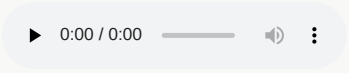
Camera shutter



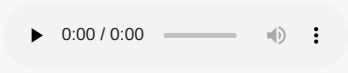
Error



Basket



Dijana



FINALLY

Six more smart solutions

Small things that save time and nerves every day.

Key cartridge

Your computer in your pocket

HOW

Profile, settings and files on an encrypted cartridge. Insert it into any Neo, at school or in a library, and you get your own desktop.

AFTERWARDS

Take out the cartridge and nothing remains on that computer.

Neo Passport

Moving to a new computer in ten minutes

MOVING HOUSE

All apps, settings and files move from the old Neo to the new one over the network or a cartridge.

NO LOSS

The old computer stays untouched until you confirm that everything has arrived.

What happened?

The system log in plain language

OVERVIEW

“Yesterday at 14:10 Wave was installed. At 18:30 Lock stopped Contract.pdf from being sent.”

GOING BACK

Every event has an undo button, where possible.

Quiet hour

Maintenance while you sleep

AT NIGHT

Updates, backups and disk checks run at night or when there is surplus sun.

MORNING

In the morning you get one short message about what was done.

Phone as a remote

The phone becomes a mouse, keyboard and remote

CONNECTION

Via Link the phone becomes a remote for Cinema and Broadcast, a keyboard for Quill and a joystick for games.

CAMERA

The phone can also serve as a second Eye camera for streaming.

Ask Neo

An assistant that runs locally

HELP

Questions about the system, in plain language, with an answer that shows where to click. Runs on Neuron, without the internet.

CHOICE

Off until you turn it on yourself. The Claude integration remains separate and optional.

PATH TO A STANDARD

Neo as a national computer

A country that brings Neo into its schools and administration gets a free OS, a licence for local assembly and manufacturing, and a translation into its own language. Socialist and social democratic countries are the natural first partners, and the door stays open to all. For countries under sanctions a legal opinion is needed.

Licence

What the country gets

SOFTWARE

Neo OS and Quartet for free, under the GPL, with local translation and Cyrillic and other scripts from day one.

HARDWARE

Schematics under an open licence, help setting up a local factory, Safe keys under the control of the state itself.

Model

Who pays

PUBLIC SECTOR

Schools, hospitals, co-operatives and administration: no licence fee.

MARKET

Commercial manufacturers pay for the “Neo” name and the compatibility certificate.

Precedents

It has happened before

HISTORY

The Pravetz in Eastern Bloc schools, Spectrum clones in the USSR, the Galaksija in Yugoslavia, the BBC Micro in Britain.

TODAY

China is replacing Windows and Intel with its own Kylin OS and Loongson chips.

PATH TO A STANDARD · HARDWARE

From prototype to open standard

The goal is for Neo to become what the IBM PC and MSX were: a specification by which several factories build compatible computers. We don't reinvent what already exists. Wherever there is a world standard, Neo uses it, and makes its own standard only for what is truly new. Timelines are estimates.

Phase	When	Hardware	What becomes the standard	Who manufactures
0. Paper and emulator	year 1	A software emulator of the whole system and an FPGA development board	Neo Hardware Manual 0.9: Copernicus, blitter, Panorama and Orpheus registers, cartridge mechanics	Universities, the demoscene, enthusiasts
1. Reference board	years 1 to 2	An off-the-shelf ARM chip and FPGA on one board, cartridge slot, DE-9, Safe	Neo Base 1.0: the mandatory minimum every Neo must have	The first local run, a few thousand units for schools and clubs
2. The first Neo package	years 2 to 4	Our own chiplets on an older process (12 to 28 nm), a shared pilot run, then a real production run	Neo Full 1.0: VSA-6000, Ray, Courier and Chronos. A certificate and the “Neo compatible” logo	Two or three licensed board and case manufacturers
3. Standard	years 4 to 6	A modern process, the 500, 2000 and 4000 family, third-party AGP-X cards	Neo 2.0, backward compatible. The standard is run by the Neo consortium	National factories under licence, cards and cartridges from several companies

We adopt

World standards, unchanged

LINKS

PCIe, CXL, UCIe for chiplets, USB-C with power delivery, M.2 and NVMe, DisplayPort and HDMI, COM Express as the basis for the Satellite module.

SOFTWARE

Vulkan, UEFI and ACPI following the ARM SystemReady rules, so another OS can be booted on Neo too. zstd, LZ4 and EROFS for compression.

We make

Only what is new

CONNECTORS

The Neo cartridge (mechanics, NVMe and signature), AGP-X (PCIe ×16 + CXL + a channel for Panorama layers), the Satellite connector, the DE-9 joystick pinout.

FORMATS

Custom chip registers, the signed Chameleon core format, neopak, the Chronos snapshot and the Neo ROM layout.

Two levels

Neo Base and Neo Full

BASIS

ARM core, FPGA fabric with Chameleon, Copernicus, Panorama, Orpheus, Courier, Safe, Neo ROM, cartridge slot, privacy switches. Enough for Neo OS, Arcade and school.

FULL

Adds VSA-6000, Ray, Newton, Neuron and Chronos. Programs ask the system what is present, so they run on Base too, only slower.

Certificate

Neo test suite

CHECKING

An open set of tests: registers, retro core timing down to the clock cycle, cartridge, Renewal, Lock and switches that really cut the power.

LOGO

Whoever passes the tests and pays for the certificate may use the name “Neo”. Whoever doesn't pay may build a compatible computer under their own name.

Governance

Neo consortium

WHO

A non-profit foundation with members: manufacturers, universities, schools, user clubs and countries that adopt Neo.

HOW

Public change proposals, voting, a new version of the standard at most once every two years and ten years of backward compatibility.

Licences

Open, but protected

SCHEMATICS AND SPECIFICATION

Schematics under CERN-OHL-S, the specification under CC BY. Anyone may read, learn and build.

NAME

The “Neo” trademark and Dijana belong to the consortium, as with Arduino: open hardware, protected name.

Manufacturing

What is made at home

LOCAL

Boards, cases, keyboards, cartridges, assembly, testing, service and FPGA cores.

IMPORTED

Silicon is made in foreign chip foundries, because that is the only realistic route today. The chip design stays ours.

Independence

No single point of failure

PROCESSOR

ARM requires a licence, so the Neo standard is written above the instruction level. A RISC-V core is the licence-free fallback, and Neo Kit programs are simply recompiled.

FOUNDRIES

Chiplets on proven processes offered by several foundries, so changing foundry isn't the end of the project.

Precedents

How others succeeded

COMPUTERS

IBM PC: published documentation gave birth to the clone industry. MSX: one standard, dozens of manufacturers. Galaksija: a schematic in a magazine.

TODAY

Arduino and Raspberry Pi (open and cheap), Framework (repairable), RISC-V and OCP (a community-run standard).

PRICES

Estimated retail prices

With stable production of 100,000 or more units a year, about twice the cost of parts. Rough estimates, not tested on the market.

NEO 500

€700 – 900

A computer in a keyboard, Eye S, OS in ROM, battery and 5G optional.

NEO 2000

€1,600 – 2,000

Desktop, Eye M, large FPGA. The Satellite x86 module can be added for €250 to 400.

NEO 4000

€3,500 – 4,500

Tower, Eye Pro, built-in x86, HBM, Blu-ray. Workstation class.

Extras: Sun panel 100 W from €120 to 180, 1 kWh station from €500 to 800. A game on cartridge is about €10 to 20 more expensive than digital. The biggest risk is developing our own chiplet package, so the first generation can use an off-the-shelf ARM chip and a separate FPGA card.

COMPARISON

What each model has

A proposed configuration for the concept, not final figures.

	Neo 500	Neo 2000	Neo 4000
Case	In a keyboard	Desktop	Tower
Main CPU	ARM (Neo paket)	ARM (Neo paket)	ARM (Neo paket)
FPGA fabric	Medium, one core	Large	Large + CGRA
VSA-6000	2 chiplets	4 chiplets	4 to 8 chiplets

	Neo 500	Neo 2000	Neo 4000
Satellite x86	None, translation only	Optional module	Built in
M.2	1	2	3
PCIe 5 / AGP-X	None	1 PCIe 5 + AGP-X	2 PCIe 5 + AGP-X
Optics	External, USB	Bay, optional	Bay, built in
Cartridge	Yes	Yes	Yes
DE-9	2 at the back	2 at the back	2 front + 2 back
Network	WiFi 7	WiFi 7 + 2.5G	WiFi 7 + 10G
OS in ROM	NOR + pSLC 2 × 16 GB	NOR + pSLC 2 × 32 GB	NOR + pSLC 2 × 32 GB
Safe chip	Yes	Yes	Yes
Privacy switches	Camera, microphone, network	Camera, microphone, network	Camera, microphone, network
Eye camera	Eye S, included	Eye M, included	Eye Pro, included
Independence	Built-in battery, 5G modem	5G modem, Sun station	5G modem, Sun station
HDMI input	None	Yes	Yes